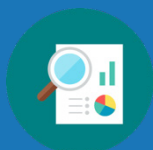


Unit 7 Overview: Trigonometric Ratios

Unit Narrative



In 8th Grade or 7th Grade Accelerated Math, students solved for missing sides of right triangles using the Pythagorean Theorem. In this unit, students will expand on this knowledge to discover and define three trigonometric ratios in order to solve real-world problems.

Section 1 provides students the opportunity to activate their prior knowledge of the Pythagorean Theorem before defining sine, cosine, and tangent. This section explores finding the opposite and adjacent sides of a right triangle when given a reference angle and using the information to form trigonometric ratios.

Section 2 builds on the trigonometric ratios defined in Section 1 and uses their understanding of similar triangles in order to find the trigonometric ratios of special right triangles. Through the discovery of the trigonometric ratios of special right triangles, students will discover the unit circle and the coordinates for different reference angles around the circle. Students will wrap up their understanding of right triangle trigonometry and the Pythagorean Theorem through real-world application problems.

Section 3 is an Honors extension of right triangle trigonometry by finding the laws of sine and cosine. Using the laws of sine and cosine, students will solve real-world application problems before using the sine ratio to find the area of triangles.

Unit At-A-Glance

Section 1: Defining Pythagorean Theorem and Trigonometric Ratios	Benchmarks Addressed	Lesson 1	The Pythagorean Theorem
	MA.912.T.1.1 MA.912.T.1.2	Lesson 2	Defining Sine and Cosine
		Lesson 3	Defining Tangent
Section 2: Using Pythagorean Theorem and Trigonometric Ratios	Benchmarks Addressed	Lesson 4	Trigonometric Ratios on the Coordinate Plane
	MA.912.T.1.1 MA.912.T.1.2	Lesson 5	Using Trigonometric Ratios
		Lesson 6	Special Right Triangles
		Lesson 7	The Unit Circle and Its Coordinates
		Lesson 8	Using Trigonometry Ratios and the Pythagorean Theorem
		Lesson 9	Using Trigonometry in the Real World
Section 3: Laws of Sine and Cosine	Benchmarks Addressed	Lesson H1	Law of Sines
	MA.912.T.1.3 (<i>Honors only</i>) MA.912.T.1.4 (<i>Honors only</i>)	Lesson H2	Law of Cosines
		Lesson H3	Area of a Triangle Using the Sine Ratio



Differentiation

Universal Design for Learning™ © CAST, 2008

Engagement <i>The “WHY” of learning</i>	• Right triangles are seen in many places throughout the real world. Consider having students document all of the different right triangles they see throughout the day, or to bring in an object that is a right triangle. This will allow students to make the real-world connection, and to become more engaged in the content. • In Lessons 2 and 3, consider having students form jigsaw groups and each member be an expert on a trigonometric function to teach to their peers.
Representation <i>The “WHAT” of learning</i>	• Provide different visual opportunities for students to access the information. Students could use shaving cream on their desk to draw right triangles for the real-world problems, or create right triangles using bricks and/or cinderblocks on a wall.
Action & Expression <i>The “HOW” of learning</i>	• Provide students an opportunity to use different online calculator software in order to access the information in different methods and avenues. • Throughout each Guided Practice and Guided Instruction, model different thought processes. Also, supplying students with checklists of items to look for when working with right triangles will help students begin to work independently.

ELL Information

This unit is a vocabulary-rich unit. Create several visual aids (chart paper sized and interactive notebook sized) of a right triangle.

- Use colored pencils and markers to identify the names of the parts. Include a visual representation of the Pythagorean Theorem showing that the sum of the squares ON the legs equals the sum of the square ON the hypotenuse.
- Consider taping a right triangle on the floor or wall and having students label the legs based on the reference angle.
- Use colored pencils and markers to accentuate the hypotenuse, the legs, the acute angles, and the legs opposite and adjacent to them. Include a three-column chart or three-circle Venn diagram to include colorful equations/formulas to illustrate the trigonometric ratios.

Stress that sketches for problems in this section should include or be RIGHT triangles. Use graph paper to improve accuracy of sketches.

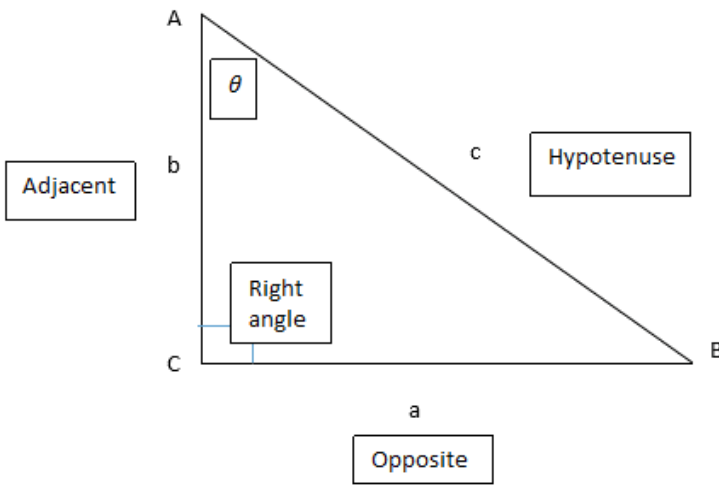
Graphic Organizers - Graphic organizers make content area information more accessible to second language learners. They convert complex information into manageable chunks. Have students use a graphic organizer to list the key steps to set up and solve problems.

Students may not be familiar with using the calculator or the steps, so provide videos and/or examples to re-watch explaining how to use the trigonometric functions of a calculator.

All glossary terms are available in multiple languages, including English, Spanish, Haitian Creole, and American Sign Language, as support for ELL students. Consider pairing the following videos with strategies from "Additional Differentiated Support" below:

- Congruent
- Hypotenuse
- Similarity

Additional Differentiated Support

Scaffolding Resources	On-Ramp
Example Activity: Provide students with index cards labeled "right angle," "hypotenuse," "adjacent," "theta," and "opposite." Provide students with a paper that has a premade right triangle already on it. Have students match and identify the areas of the right triangle using the given index cards (understand that the right angle and the hypotenuse never change, but theta can be either of the other angles, and then the adjacent and opposite would correspond to theta).  Example activity: Provide students with a right triangle that has two of the angle measurements given. Have students determine the measurement of the missing angle. For instance, if the given measurements are 90 degrees and 35 degrees, the missing measurement is 55 degrees because the three angle measurements must equal 180 degrees.	Use the videos and practice problems in the On-Ramp to Geometry personalized learning tool to review the following topics for this unit. Videos are available in English and Spanish. • Pythagorean Theorem • Area of a Triangle



Section Coherence

Section 1: Defining Pythagorean Theorem and Trigonometric Ratios (Lessons 1-3)

Benchmarks Addressed	Learning Targets	
MA.912.T.1.1 Define trigonometric ratios for acute angles in right triangles. - Clarification 1: Instruction includes using the Pythagorean Theorem and using similar triangles to demonstrate that trigonometric ratios stay the same for similar right triangles. - Clarification 2: Within the Geometry course, instruction includes using the coordinate plane to make connections to the unit circle. - Clarification 3: Within the Geometry course, trigonometric ratios are limited to sine, cosine and tangent. MA.912.T.1.2 Solve mathematical and real-world problems involving right triangles using trigonometric ratios and the Pythagorean Theorem. - Clarification 1: Instruction includes procedural fluency with the relationships of side lengths in special right triangles having angle measures of $30^\circ - 60^\circ - 90^\circ$ and $45^\circ - 45^\circ - 90^\circ$.	Lesson 1	I can find the missing side of a right triangle using the Pythagorean Theorem.
	Lesson 2	I can define sine and cosine using similar triangles.
	Lesson 3	I can use the ratios of sine and cosine to define the tangent ratio.

Section 2: Using Pythagorean Theorem and Trigonometric Ratios (Lessons 4-9)

Benchmarks Addressed	Learning Targets	
MA.912.T.1.1 Define trigonometric ratios for acute angles in right triangles. - Clarification 1: Instruction includes using the Pythagorean Theorem and using similar triangles to demonstrate that trigonometric ratios stay the same for similar right triangles. - Clarification 2: Within the Geometry course, instruction includes using the coordinate plane to make connections to the unit circle. - Clarification 3: Within the Geometry course, trigonometric ratios are limited to sine, cosine and tangent. MA.912.T.1.2 Solve mathematical and real-world problems involving right triangles using trigonometric ratios and the Pythagorean Theorem. - Clarification 1: Instruction includes procedural fluency with the relationships of side lengths in special right triangles having angle measures of $30^\circ - 60^\circ - 90^\circ$ and $45^\circ - 45^\circ - 90^\circ$.	Lesson 4	I can determine trigonometric ratios using the coordinate plane.
	Lesson 5	- I can use trigonometric ratios to find the side of a right triangle. - I can use trigonometric ratios to find an angle of a right triangle.
	Lesson 6	- I can use side ratios of 45–45–90 triangles to solve problems. - I can use side ratios of 30–60–90 triangles to solve problems.
	Lesson 7	- I can explore relationships between the coordinates of vertices when the triangles are in different quadrants. - I can use side ratios of 45–45–90 triangles to solve problems. - I can use side ratios of 30–60–90 triangles to solve problems.

	Lesson 8	- I can use trigonometric ratios to solve mathematical problems. - I can use the Pythagorean Theorem to solve mathematical problems.
	Lesson 9	I can solve real-world problems using trigonometric ratios and the Pythagorean Theorem.

Section 3: Laws of Sine and Cosine (Lessons H1-H3)

Benchmarks Addressed	Learning Targets	
MA.912.T.1.3 (Honors only) Apply the Law of Sines and the Law of Cosines to solve mathematical and real-world problems involving triangles. MA.912.T.1.4 (Honors only) Solve mathematical problems involving finding the area of a triangle given two sides and the included angle. <i>Clarification 1: Problems include right triangles, heights inside of a triangle and heights outside of a triangle.</i>	Lesson H1	I can solve mathematical and real-world problems involving triangles using the Law of Sines.
	Lesson H2	I can solve mathematical and real-world problems that involve triangles using the Law of Cosines.
	Lesson H3	I can find the area of a triangle using the sine ratio.



Vertical Alignment

Grade 8 Unit 5
Inequalities and
Pythagorean Theorem



Geometry Unit 7
Trigonometric Ratios

Prior Course(s)

MA.8.GR.1.1 Apply the Pythagorean Theorem to solve mathematical and real-world problems involving unknown side lengths in right triangles.

MA.8.GR.1.2 Apply the Pythagorean Theorem to solve mathematical and real-world problems involving the distance between two points in a coordinate plane.

Course Level

MA.912.T.1.1 Define trigonometric ratios for acute angles in right triangles.

MA.912.T.1.2 Solve mathematical and real-world problems involving right triangles using trigonometric ratios and the Pythagorean Theorem.

MA.912.T.1.3 (*Honors only*) Apply the Law of Sines and the Law of Cosines to solve mathematical and real-world problems involving triangles.

MA.912.T.1.4 (*Honors only*) Solve mathematical problems involving finding the area of a triangle, given two sides and the included angle.

7.1 The Pythagorean Theorem

Lesson Introduction

 Benchmarks	MA.912.T.1.1 Define trigonometric ratios for acute angles in right triangles. MA.912.T.1.2 Solve mathematical and real-world problems involving right triangles using trigonometric ratios and the Pythagorean Theorem.		 Learning Target	I can find the missing side of a right triangle using the Pythagorean Theorem.
 Benchmark Coherence	Building On MA.8.F.1.1 Given a set of ordered pairs, a table, a graph, or a mapping diagram, determine whether the relationship is a function. Identify the domain and range of the relation.		Working Toward N/A	
 Essential Questions	- How do you find the missing side of a right triangle? - What is the relationship between the sides of a right triangle?			
 Lesson Narrative	This lesson allows students to work together to explore the relationship between the sides of a right triangle. Using the information found in the Collaboration, students will work through practice problems over the Pythagorean Theorem. Students will then complete Guided Practice problems and individual practice over Pythagorean Theorem.			
 Component Summary	7.1.1 Warm-Up (~5 min.)	Discussing success from failure as told by a Sales Strategist (<i>SE p. 369</i>)	7.1.3 Guided Practice (15-20 min.)	Solving for missing sides of a right triangle (<i>SE p. 372</i>)
	7.1.2 Collaboration (10-15 min.)	Exploring the Pythagorean Theorem (<i>SE p. 370</i>)	7.1.4 Your Turn! (~10-15 min.)	Applying Pythagorean Theorem to find the missing side length of a right triangle (<i>SE p. 373</i>)
	7.1.5 Wrap-up (~5 min.)	Students summarize their learning (<i>SE p. 374</i>)		
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> - N/A <u>Additional Terms:</u> - Hypotenuse - Pythagorean Theorem		- Failure to Find Success Video - Graphing utility (Optional): Index cards or copy paper, colored pencils, scissors	7.1.1 Warm-Up contains a video to accompany the question prompts

 Teacher Tips	Common Misconceptions - Students may struggle with differentiating between the legs and hypotenuse of right triangles. - Students may believe the hypotenuse, or c, will always be the missing side length, still finding the sum of a and c rather than the difference in those cases.	Things to Consider Consider allowing students to cut out copies of the squares in 7.1.2 Collaboration to find the connection between the area of the squares forming the legs, and the area of the square forming the hypotenuse.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Notice and Wonder At the start of the Collaboration during 7.1.2 Collaboration, have students discuss what they notice and wonder. This allows students to begin asking questions and exploring the Pythagorean Theorem before working with the Pythagorean Theorem, making informal observations that ultimately connect to formalized concepts.	MA.K12.MTR.3.1: Mathematical Fluency Allow students multiple opportunities to solve Pythagorean Theorem problems that are presented in different ways. Allow students time to discuss their way of solving with a partner and provide feedback for students.
 Differentiate Instruction	In 7.1.2 Collaboration, students have the opportunity to explore one real-world application of the Pythagorean Theorem by investigating the shorter distance between two different walking paths from the beach. Consider this scenario as an opportunity to differentiate by content to engage your students in ways unique to your contexts. - Provide an example from your school campus or local community that students would be familiar with and may agree or disagree based on their personal experience. - Prompt students to create their own example based on a location they are familiar with or walk often. Challenge them to find actual distances before or after class to investigate or verify their findings mathematically.	
Lesson Notes:		

7.1.1 Warm-Up: Sales Strategist

Component Overview: Students will look at how to respond to failure after watching a short video.



7.1.1 Warm-Up: Sales Strategist

Kishan Patel says that he was never really a good student in high school. He graduated with a 2.9 GPA. All of those challenges prepared him for business. Business is about failing and doing better. Every time you fail, you have to improve in some way to get to the next level.

1. Based on the statement above, how do you think Kishan views failure?

Answers will vary.

2. When you fail at something, describe how you typically react.

Answers will vary.



Consider allowing a few students to share their responses. Do not spend too much time on this topic, but instead allow it to be used to explore Growth Mindset.

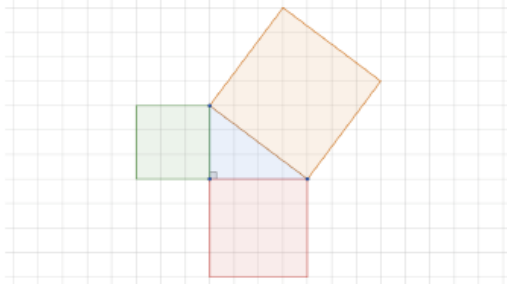
7.1.2 Collaboration: Pythagorean Theorem

Component Overview: Students will work in pairs or groups to explore the relationship between the legs and hypotenuse of a right triangle.



7.1.2 Collaboration: Pythagorean Theorem

A right triangle is shown with three squares.

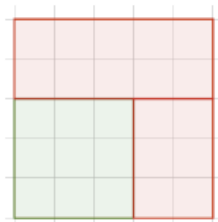


- The three squares from the diagram are shown. Complete the table with the area of each square.

Side Length (units)	Area of Square (square units)	
3	9	
4	16	
5	25	

- Explain how the diagram shows that the area of the green square plus the area of the red square is equal to the area of the orange square.

The number of unit squares in both the green and red squares make up exactly the same number of unit squares in the orange square.



For a kinesthetic activity, have students use paper or index cards to cut out each square for this activity. Students can use the cut outs to explore the relationship between the orange square and the red and green squares.



Notice and Wonder

Before students answer questions, consider using this instructional routine to allow students to write down things they notice and wonder about the image.

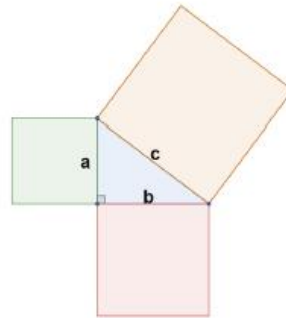
- What do you notice about the image?
- What do you wonder about the image?

7.1.2 Collaboration: Pythagorean Theorem (continued)

3. Complete the relationship of the areas by filling in the box.

$$3^2 + 4^2 \boxed{=} 5^2$$

4. Use the diagram to write an explanation of the Pythagorean Theorem, $a^2 + b^2 = c^2$.
Sides a and b are legs of the right triangle. Side c is the hypotenuse. If the lengths of these two sides, a and b , are squared and then added together, their sum will be equal to the length of the hypotenuse, c , squared.



5. Iverson and Jake are at the beach and plan to meet a friend for lunch.



Iverson decided to walk directly to the restaurant, while Jake decides to take a route that will allow him to cross the street using the crosswalk. The diagram shows the two routes. Jake walks horizontally 252 meters (m) and then vertically 115 m.



Think-Pair-Share

Provide students an opportunity to share their group discussions with other groups in order to discuss different opinions.



Utilize inquiry for students to make connections between the visual squares and the Pythagorean Theorem:

- What is the relationship between the green/red squares and the orange square?
- Would that always be true? Verify.
- What does that make you think about the relationship between the two legs of a right triangle and their hypotenuse?



Students who missed the 8th grade benchmarks may not know the definition of hypotenuse.

7.1.2 Collaboration: Pythagorean Theorem (continued)



a. Complete the table.

	Iverson	Jake
Distance Walked	277 m	367 m

b. Discuss with your partner the advantages and disadvantages of each person's walk to the restaurant. Summarize your discussion in the table.

Answers may vary.

Person	Advantages	Disadvantages
Iverson	<i>Shorter distance.</i>	<i>Not using the crosswalk which is illegal to cross a road.</i>
Jake	<i>Using the crosswalk which is the legal way to go across a street.</i>	<i>Longer distance.</i>



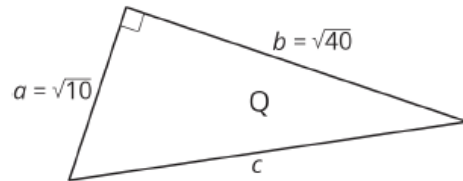
Which way do you think is the quickest way to get to the lunch spot?



When would it be better to take the longer path versus the direct path?

7.1.2 Collaboration: Pythagorean Theorem (continued)

6. A right triangle is shown with side lengths $\sqrt{10}$ and $\sqrt{40}$ units.



Find the length of the hypotenuse using the Pythagorean Theorem. If the measure of the side length is an irrational value, write an exact solution and an approximate solution.

- | | |
|------------------------------------|-----------------------------------|
| a. State Pythagorean Theorem | $a^2 + b^2 = c^2$ |
| b. Substitute | $\sqrt{10}^2 + \sqrt{40}^2 = c^2$ |
| c. Solve for the missing variable. | $10 + 40 = c^2$ |
| d. Solutions | $50 = c^2$ |

Exact solution:
 $c = \sqrt{50}$

Approximate answer:

$$c \approx 7.071$$



- Why does $\sqrt{277^2} = 277$, and $\sqrt{367^2} = 367$?
- What does that make you think about what your final answer for the hypotenuse might look like?

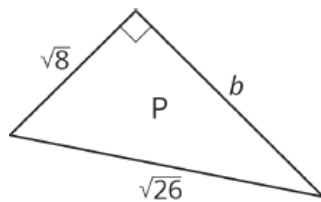
7.1.3 Guided Practice: Pythagorean Theorem

Component Overview: Students use the Pythagorean Theorem to find the missing side lengths of right triangles.



7.1.3 Guided Practice: Pythagorean Theorem

- Find the exact measure of b in the diagram.



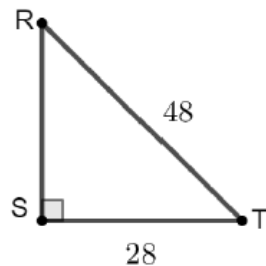
$$\sqrt{8}^2 + b^2 = \sqrt{26}^2$$

$$8 + b^2 = 26$$

$$b^2 = 18$$

$$b = \sqrt{18}$$

- $\triangle RST$ is shown, where $ST = 28$ and $RT = 48$.



Find the length of $\overline{SR}$ to the nearest thousandth.

$$(SR)^2 + 28^2 = 48^2$$

$$(SR)^2 + 784 = 2304$$

$$(SR)^2 = 1520$$

$$SR = 38.987$$



Use observation data from the 7.1.2 Collaboration to determine how much guidance is needed to bring together key ideas in this Guided Practice. If necessary, consider modeling question 1 to address common misconceptions shown in the Collaboration.



Students may think the missing side is always the c -value in the equation. Consider having students draw a line or arrow from the right angle pointing towards the hypotenuse, in order to find c .



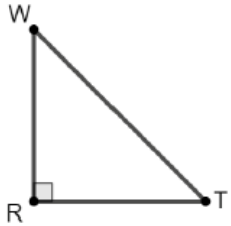
In question 2, students may find that $SR = 20$ if they subtract 48 and 28 before squaring. Facilitate a quick discussion about how to use order of operations when solving an equation.



Consider allowing students to use a calculator for question 2 and question 3. Determine what place value you would like your students to round to for their final answer.

7.1.3 Guided Practice: Pythagorean Theorem (continued)

3. $\triangle WRT$ is shown, where $WT = 18.4$ and $RT = 12.1$.



Find the length, to the nearest thousandth, of $\overline{WR}$.

$$(WR)^2 + 12.1^2 = 18.4^2$$

$$WR = 13.862$$



This problem provides an opportunity to determine a student's understanding of solving algebraic equations involving squares and square roots.

Students who struggle with radicals would benefit from a review of solving equations using Algebra 1 On-Ramp "Simplifying Square Roots," and/or "Approximating Irrational Numbers."

7.1.4 Your Turn!

Component Overview: Students practice using the Pythagorean Theorem to find the missing side lengths of right triangles.



7.1.4 Your Turn!

1. In $\triangle HMX$, $m\angle MXH = 90^\circ$, $MH = 12$, and $MX = 7$. Determine the length of $\overline{XH}$. Round to the nearest thousandth.

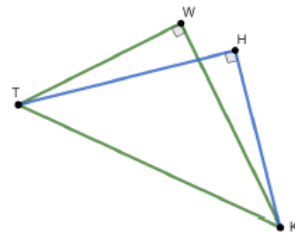
$$7^2 + (XH)^2 = 12^2$$

$$49 + (XH)^2 = 144$$

$$(XH)^2 = 95$$

$$XH = 9.747$$

2. $\triangle TWK$ and $\triangle THK$ are shown, where $TW = \sqrt{45}$, $TH = \sqrt{68}$, and $TK = \sqrt{113}$. Find the length of $\overline{WK}$ and $\overline{HK}$. Round to the nearest thousandth.



$$(WK)^2 + \sqrt{45}^2 = \sqrt{113}^2$$

$$WK = 8.246$$

$$(HK)^2 + \sqrt{68}^2 = \sqrt{113}^2$$

$$HK = 6.708$$



The information provided from the student here, coupled with what was observed during instruction, can provide information for differentiating instruction in later lessons.

7.1.5 Wrap-Up: Cool Down

Component Overview: Students write a short summary of their learning in the lesson. Student responses can be used to identify misconceptions and plan for upcoming instruction.



7.1.5 Wrap-Up: Cool Down

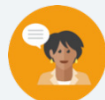
1. Write an explanation to a classmate who might have missed today's lesson, in your own words, about the Pythagorean Theorem.

Answers will vary.

7.2 Defining Sine and Cosine

Lesson Introduction

 Benchmarks	MA.912.T.1.1 Define trigonometric ratios for acute angles in right triangles. MA.912.T.1.2 Solve mathematical and real-world problems involving right triangles using trigonometric ratios and the Pythagorean Theorem.		 Learning Target	I can define sine and cosine using similar triangles.
 Benchmark Coherence	Building On: MA.8.GR.1.1 Apply the Pythagorean Theorem to solve mathematical and real-world problems involving unknown side lengths in right triangles.		Working Toward: N/A	
 Essential Questions	- How do I identify the opposite and adjacent sides in a right triangle? - What is a reference angle? - What are trigonometric ratios?			
 Lesson Narrative	Through exploration, students will determine that the ratio of the rise and run of similar right triangles will always be the same. Students will work together to discover the ratios involving the opposite, adjacent, and hypotenuse sides of right triangles, being formally introduced to the sine, cosine, and tangent of given reference angles.			
 Component Summary	7.2.1 Warm-Up (~5 min.)	Determine which region is a larger area (<i>SE p. 375</i>)	7.2.4 Bring It Together (5 min.)	Formalize the trigonometric ratios (<i>SE p. 379</i>)
	7.2.2 Exploration (10-15 min.)	Determine the relationships of slopes in similar right triangles (<i>SE p. 375</i>)	7.2.5 Guided Practice (5 min.)	Write trigonometric ratios given a reference angle (<i>SE p. 380</i>)
	7.2.3 Collaboration (10-15 min.)	Label the sides of a right triangle (<i>SE p. 378</i>)	7.2.6 Wrap-Up (~5 min.)	Write a letter to a friend describing the day's lesson (<i>SE p. 381</i>)
 Resources	Glossary		Materials	
	Key Terms: - Sine - Cosine - Tangent Additional terms: - Opposite leg - Adjacent leg - Hypotenuse - Slope		- Straightedge - Protractor (Optional) Colored pencils (Optional) Tape	
		Additional Preparation - This is a vocabulary-rich lesson. Consider creating several right triangles taped on the floor and/or wall for students to explore the vocabulary. - Consider taping a large right triangle on the floor of the classroom prior to the lesson for kinesthetic learners.		

 Teacher Tips	Common Misconceptions	Things to Consider
	- During 7.2.3 Collaboration, students may believe only two of the angles are congruent. - Beginning in 7.2.3 Collaboration and throughout the lesson, students may mislabel the sides based on the reference angle.	- Make sure students are exposed to several different orientations of a right triangle, so that they are not only considering right triangles positioned in the same direction. - Use colored pencils to color code the sides when labeling triangles based on the reference angle.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Collect and Display During 7.2.2 Exploration, listen for and collect responses about the slopes of special right triangles. Facilitate a discussion with students about the different conjectures.	MA.K12.MTR.5.1: Patterns and Structure Throughout the lesson, students will be working to find patterns in the relationships. Students will first build relationships from patterns for 30-60-90 and 45-45-90 triangles. Then students will work with reference angles and ratios to discover the relationships between the ratios for sine, cosine, and tangent.
 Differentiate Instruction	An alternative way of reviewing content and working through 7.2.2 Exploration would be to have students work through 3 or 4 stations. - Station 1: On Ramp - Determine slope from a graph. - Station 2: 30-60-90 triangles - Students complete the first part of the exploration task. - Station 3: 45-45-90 triangles - Students complete the second part of the exploration task. (Optional) Station 4: Remediation or re-teaching the previous lesson based on data collected. Additional differentiation opportunities are denoted throughout the lesson guide.	
Lesson Notes:		

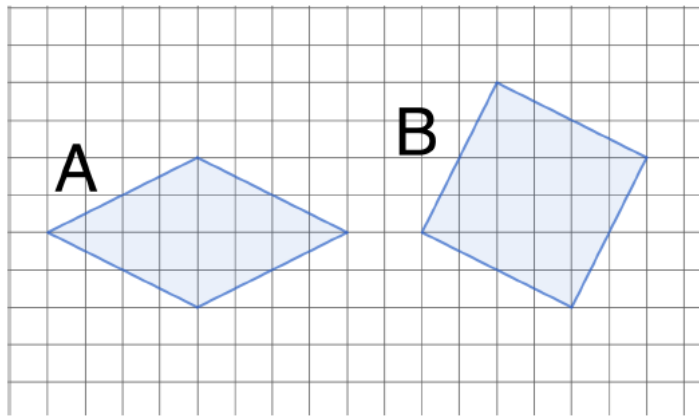
7.2.1 Warm-Up: Which Region Is Larger?

Component Overview: Students are determining which shaded region has a larger area and explaining their reasoning.



7.2.1 Warm-Up: Which Region Is Larger?

1. Two grids are shown with regions shaded.



- a. Which shaded region is larger?
Figure B has the larger shaded region.
- b. Explain your reasoning.
Answers will vary.



Watch to see how students reason about which area is larger. Do they try to use diagonal distances as equivalent to one gridline? Do they imagine decomposing the shaded regions to determine the number of whole squares?

7.2.2 Exploration: Similar Right Triangles

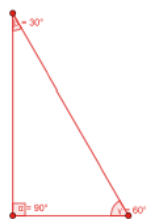
Component Overview: Students are exploring the slopes of the hypotenuses in 30-60-90 and 45-45-90 triangles.



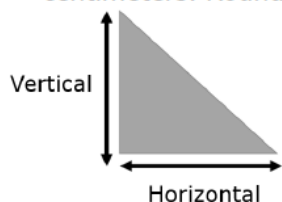
7.2.2 Exploration: Similar Right Triangles

- Using the space below, create a right triangle that has one angle with a measure of 60° . Be accurate with measurements.

Answers will vary.



- Measure the horizontal and vertical distance of your triangle. Determine the slope of your triangle in centimeters. Round to the nearest tenth.



Vertical	Horizontal	Slope ($\frac{\text{vertical}}{\text{horizontal}}$)
7.5	4.4	$\frac{7.5}{4.4} \approx 1.7$

Answers will vary.

- Record the slopes from two of your classmates. You are the only person who should write in the first two columns of the table. Have the person initial stating that your slope is correct.

Answers will vary.

Classmate	Slope	Initials

- What do you notice about the slopes?

Answers will vary. They are all approximately equal to 1.7.



An alternative way of reviewing content and working through the exploration activity would be to have students work through 3 stations:

Station 1: On Ramp - Determine Slope from a Graph

Station 2: 30-60-90 - Students complete the first part of the Exploration task.

Station 3: 45-45-90 - Students complete the second part of the Exploration task.

A 4th station could be utilized for remediation or re-teaching of the previous day's lesson.



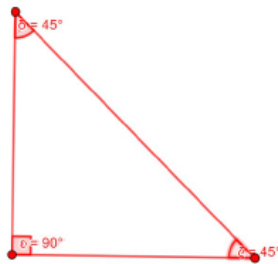
Have students create a variety of different triangles using GeoGebra as they complete the exploration.

7.2.2 Exploration: Similar Right Triangles (continued)

b. What do you wonder?

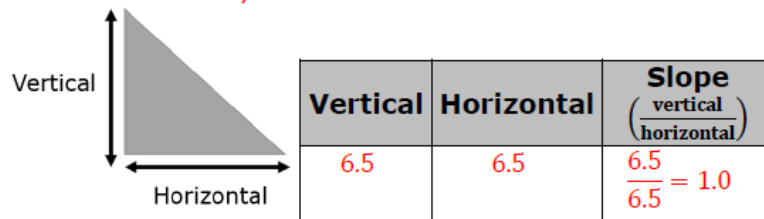
Answers will vary. Is this true for all triangles that have a 60° angle at its base? Will all triangles with interior angles of 30° , 60° , and 90° , always have the same approximate slope?

4. Draw a right triangle that has an angle of 45° . Be accurate with measurements.



5. Measure the horizontal and vertical distance of your triangle. Determine the slope of your triangle in centimeters. Round to the nearest tenth.

Answers will vary.



6. Record the slopes from two of your classmates. *You are the only person who should write in the second column of the table.* Have the person initial stating that your slope is correct.

Answers will vary.

Classmate	Slope	Initials



How are the slopes similar if the 30- and 60-degree angles swap locations? How are they different?



Consider allowing students to write their notices and wonderings on two sticky notes and place them together in an area of the classroom for everyone to view. Another alternative is to use an online whiteboard service or forum space for students to post their responses.

7.2.2 Exploration: Similar Right Triangles (continued)

- a. What do you notice about the slopes?

Answers will vary. They are all approximately equal to 1.

- b. What do you wonder?

Answers will vary. Is this true for all triangles with a base of 45° ?

7. Ask two classmates for the "Notice and Wonder" what they wrote for both triangles. Record their responses below. *You are the only person who should write in the first three columns of the table.* Have the person initial that it is written correctly.

Answers will vary.

Classmate	Notice	Wonder	Initials
Classmate #1	<i>The right triangles we constructed with a base of 60° have the same approximate slope of 1.7.</i>	<i>Is this true for all right triangles with a base of 60°?</i>	
Classmate #2	<i>The right triangles we constructed with a base of 45° all have the same approximate slope of 1.0.</i>	<i>Is this true of all right triangles with a base of 45°?</i>	

8. Write a conjecture for each of the right triangles that were drawn.

Angle Measures	Conjecture
$30^\circ - 60^\circ - 90^\circ$	<i>Answers will vary. The slope of a $30^\circ - 60^\circ - 90^\circ$ triangle is approximately 1.7.</i>
$45^\circ - 45^\circ - 90^\circ$	<i>Answers will vary. The slope of a $45^\circ - 45^\circ - 90^\circ$ triangle is equal to 1.</i>



Does it matter which angle is used when finding the slope? Justify your response.



In future lessons, students will be working with 30° - 60° - 90° and 45° - 45° - 90° triangles. Without giving students the special right triangle rules, allow them to discuss anything they may have noticed or wondered about the triangles as they were completing the exploration activity.

7.2.2 Exploration: Similar Right Triangles (continued)

9. The triangles drawn by you and your classmates during the Exploration vary. Using either the triangles explored in Questions 1-3 or Questions 4-6, explain why these triangles are similar.

Each of the triangles that were drawn had two congruent angles, so all of the triangles explored in Questions 1-3 are similar by AA similarity and all of the triangles explored in Questions 4-6 are similar by AA similarity.



Right triangles with one other congruent angle are similar because of AA similarity.



Collect and Display

Listen for and collect responses about the slopes of special right triangles. Facilitate a discussion with students about the different conjectures.

7.2.3 Collaboration: Introduction to Trigonometric Ratios

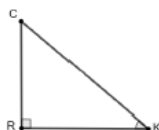
Component Overview: Students are discovering how to label the opposite and adjacent sides based on the reference angle. Students will then explore the relationship between sine, cosine, and tangent in similar triangles.



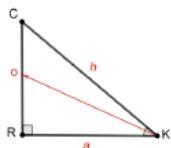
7.2.3 Collaboration: Introduction to Trigonometric Ratios

Work with your partner to complete the following.

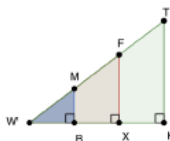
1. $\triangle CRK$ is shown, where $m\angle CRK = 90^\circ$ and $\angle CKR$ is marked as a reference angle.



- a. Draw an arrow from $\angle CKR$ to side $\overline{CR}$. Label that side with an o for opposite.
- b. Label side $\overline{RK}$ with an a for adjacent.
- c. Label the hypotenuse, $\overline{CK}$, with an h for hypotenuse.



2. The diagram shows three right triangles that share the reference $\angle W$. Explain why $\triangle WMB$, $\triangle WFX$, and $\triangle WTK$ are similar.



Answers will vary.

All three triangles have a right angle and all three share $\angle W$, therefore they are all similar by AA similarity postulate, which states that two or more triangles are similar if two angles in one triangle are congruent to two angles in another triangle.



For a kinesthetic approach for the lesson, allow students to use colored pencils to color code each side. Another option is to tape a right triangle to the floor, and have students walk to the opposite leg and reach down to touch the adjacent leg.



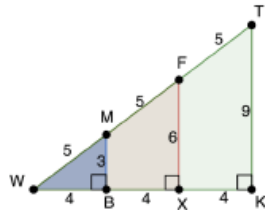
- Which side would be the opposite if $\angle KCR$ was the reference angle?
- Which side would be the adjacent side?
- Why would the sides change if we used a different reference angle?

7.2.3 Collaboration: Introduction to Trigonometric Ratios (continued)

3. Complete the table.

Triangle	Leg Opposite $\angle W$	Leg Adjacent to $\angle W$	Hypotenuse
$\triangle WMB$	$\overline{MB}$	$\overline{WB}$	$\overline{WM}$
$\triangle WFX$	$\overline{FX}$	$\overline{WX}$	$\overline{WF}$
$\triangle WTK$	$\overline{TK}$	$\overline{WK}$	$\overline{WT}$

4. The diagram shows measurements for the three triangles.



Complete the table using angle $\angle W$ as the reference angle.

Triangle	$\frac{\text{opposite}}{\text{hypotenuse}}$	$\frac{\text{adjacent}}{\text{hypotenuse}}$	$\frac{\text{opposite}}{\text{adjacent}}$
$\triangle WMB$	$\frac{3}{5}$	$\frac{4}{5}$	$\frac{3}{4}$
$\triangle WFX$	$\frac{6}{10} = \frac{3}{5}$	$\frac{8}{10} = \frac{4}{5}$	$\frac{6}{8} = \frac{3}{4}$
$\triangle WTK$	$\frac{9}{15} = \frac{3}{5}$	$\frac{12}{15} = \frac{4}{5}$	$\frac{9}{12} = \frac{3}{4}$



Common Misconception

Students may believe only two of the angles are congruent. Facilitate a discussion about how the sum of a triangle is 180, and if 2 angles are congruent, then the 3rd angle has to be congruent.



For a visual and kinesthetic approach, consider having students trace the three triangles on patty paper. Students can label side lengths and mark congruent sides.

7.2.4 Bring It Together: Trigonometric Ratios

Component Overview: Using the information from 7.2.2 Exploration and 7.2.3 Collaboration, students will bring together all their findings to define sine, cosine, and tangent.



7.2.4 Bring It Together: Trigonometric Ratios

Since $\triangle WMB$, $\triangle WFX$, and $\triangle WTK$ are similar, the $\frac{\text{opposite}}{\text{hypotenuse}}$ ratios are equivalent to each other. This is also true for the ratios $\frac{\text{adjacent}}{\text{hypotenuse}}$ and $\frac{\text{opposite}}{\text{adjacent}}$. For this reason, it is possible to define these ratios in any right triangle.

The ratios use the reference angle theta (θ).



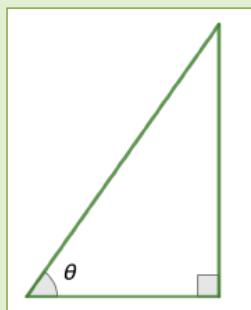
Facilitate a discussion over the importance of knowing the reference angle when using sine, cosine, and tangent.

Trigonometric Ratios

$$\sin(\theta) = \frac{\text{opposite}}{\text{hypotenuse}}$$

$$\cos(\theta) = \frac{\text{adjacent}}{\text{hypotenuse}}$$

$$\tan(\theta) = \frac{\text{opposite}}{\text{adjacent}}$$



7.2.5 Guided Practice: Trigonometric Ratios

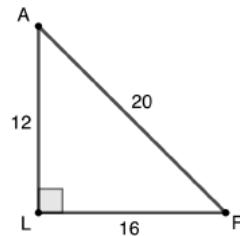
Component Overview: Students are determining the ratio of sides in right triangles when given a reference angle.



7.2.5 Guided Practice: Trigonometric Ratios

1. $\triangle FLA$ is shown. $m\angle ALF = 90^\circ$, $AL = 12$, $LF = 16$, and $AF = 20$.

Complete the table using each angle as the reference angle.

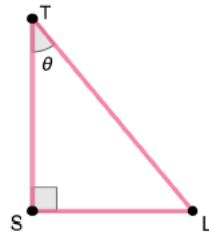


Angle	$\frac{\text{opposite}}{\text{hypotenuse}}$	$\frac{\text{adjacent}}{\text{hypotenuse}}$	$\frac{\text{opposite}}{\text{adjacent}}$
$\angle F$	$\frac{12}{20} = \frac{3}{5}$	$\frac{16}{20} = \frac{4}{5}$	$\frac{12}{16} = \frac{3}{4}$
$\angle A$	$\frac{16}{20} = \frac{4}{5}$	$\frac{12}{20} = \frac{3}{5}$	$\frac{16}{12} = \frac{4}{3}$

2. $\triangle TLS$ is shown where $LS = \sqrt{35}$ and $TS = \sqrt{42}$.
a. Determine the length of TL .

$$(\sqrt{35})^2 + (\sqrt{42})^2 = (TL)^2$$

$$TL = \sqrt{77}$$



- b. Complete the table for the reference angle θ .

Angle	$\frac{\text{opposite}}{\text{hypotenuse}}$	$\frac{\text{adjacent}}{\text{hypotenuse}}$	$\frac{\text{opposite}}{\text{adjacent}}$
θ	$\frac{\sqrt{35}}{\sqrt{77}}$	$\frac{\sqrt{42}}{\sqrt{77}}$	$\frac{\sqrt{35}}{\sqrt{42}}$



Common Misconception

Students may mislabel the sides based on the reference angle. Allow students to use a colored pencil to color code the sides, or to redraw the triangle to find the sides for the second reference angle.



Common Misconception

If students determine the length is 77, then they did not take the square root of c^2 . Facilitate a discussion over inverse operations in order to discuss how to find c .



Consider challenging students to discover the relationship between the ratios of sine and cosine to the ratio of tangent.

- How can you use the ratios for sine and cosine to find the ratio for tangent?



7.2.6 Wrap-Up: Cool Down

Component Overview: Students complete a letter to a friend explaining the day's lesson.



7.2.6 Wrap-Up: Cool Down

1. Your friend is absent today and will need the notes from class. Write a quick note to explain what was missed in today's lesson.

Answers will vary.

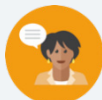


What would a note look like for you as a teacher? Consider your own letter and the vocabulary you would expect to see in the letter. Student responses will also give you insight to the level of understanding of the lesson.

7.3 Defining Tangent

Lesson Introduction

Lesson Introduction				
 Benchmarks	MA.912.T.1.1 Define trigonometric ratios for acute angles in right triangles. MA.912.T.1.2 Solve mathematical and real-world problems involving right triangles using trigonometric ratios and the Pythagorean Theorem.		 Learning Target	I can use the ratios of sine and cosine to define the tangent ratio.
 Benchmark Coherence	Building On MA.8.GR.1.1 Apply the Pythagorean Theorem to solve mathematical and real-world problems involving unknown side lengths in right triangles.		Working Toward N/A	
 Essential Questions	- How do sine and cosine define tangent? - How can I identify trigonometric ratios when given a reference angle? - How do I find missing sides of a right triangle?			
 Lesson Narrative	Through exploration, students will examine the relationship between sine and cosine to develop the tangent ratio. The tangent ratio is formally defined as ratio of $\frac{\text{sine}}{\text{cosine}}$ as well as $\frac{\text{opposite leg}}{\text{adjacent leg}}$. Students will find missing side lengths and identify the ratios of sine, cosine, and tangent.			
 Component Summary	7.3.1 Warm-Up (~5 min.)	Discuss notices and wonderings from a given image (<i>SE p. 382</i>)	7.3.3 Guided Instruction (10 min.)	Identify sine, cosine, and tangent when given a reference angle (<i>SE p. 384</i>)
	7.3.2 Exploration (15-20 min.)	Explore the relationship of sine and cosine to find tangent (<i>SE p. 382</i>)	7.3.4 Guided Practice (5-10 min.)	Solve for missing side lengths and identifying the ratios of sine, cosine, and tangent (<i>SE p. 386</i>)
	7.3.5 Wrap-Up (~5 min.)	Students summarize their learning (<i>SE p. 387</i>)		
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> - Tangent <u>Additional Terms:</u> - Sine - Cosine - Reference angle - Adjacent leg - Opposite leg - Hypotenuse		(Optional) Calculator (Optional) Index Cards (Optional) Right triangles with no angles or sides labeled (Optional) Colored pencils	- Index cards with sine, cosine, and tangent written on them can provide support to kinesthetic learners when multiplying sine and cosine. - Consider providing copies of right triangles with no angles or sides labeled for students to use when no image is presented.

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may multiply the numerators only when multiplying fractions. - Students may mislabel the opposite leg and the adjacent leg. - Students may forget to square the coefficient when squaring a radical expression.	Consider creating additional practice with right triangles that are not positioned with the right angle located at the end of a horizontal leg on the page.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Algebra Talk During 7.3.4 Guided Practice, allow students to share their different drawings and discuss why the drawings are correct. Use the opportunity to discuss how right triangles can be drawn differently but must always have a right angle, and how the hypotenuse is the longest side.	MA.K12.MTR.6.1: Reasonableness Throughout the lesson, discuss how the different representations of the ratio of sine, cosine, and tangent are correct. Facilitate a discussion with students about how simplifying radical expressions leads to equivalent ratios. Encourage students to check to make sure their answers are reasonable and correct.
 Differentiate Instruction	In 7.3.2 Exploration, allow students to color code the sides of the triangle in order to identify opposite, adjacent, and hypotenuse. Consider creating flashcards for students to use for a kinesthetic approach to multiplying and dividing fractions when discovering how sine multiplied by cosine will equal tangent. Additional differentiation opportunities are denoted throughout the lesson guide.	
Lesson Notes:		

7.3.1 Warm-Up: Notice and Wonder

Component Overview: Using the image given, students will identify what they notice and wonder about the image.



7.3.1 Warm-Up: Notice and Wonder



1. What do you wonder?

Answers will vary.

Why is the faucet in the air?

How does the water get to the faucet?

2. What do you notice?

Answers will vary.

Faucet is in the air.

Water is pouring out.



Consider allowing a few students to share their responses. Do not spend too much time on this topic, but instead allow it to be used to explore growth mindset.

7.3.2 Exploration: Relating Sine, Cosine and Tangent

Component Overview: Students are exploring the relationship between sine and cosine to develop tangent.



7.3.2 Exploration: Relating Sine, Cosine and Tangent

- The table gives the sine and cosine values as decimals for different reference angle measures. Using a calculator and the values in the table, find the ratio of $\frac{\text{sine}}{\text{cosine}}$. Round to the nearest thousandth.

Angle Measure	Sine	Cosine	$\frac{\text{sine}}{\text{cosine}}$
10°	0.174	0.985	0.177
20°	0.342	0.940	0.364
30°	0.500	0.866	0.577
40°	0.643	0.766	0.839
50°	0.766	0.643	1.191
60°	0.866	0.500	1.732
70°	0.940	0.342	2.749
80°	0.985	0.174	5.661

- Use a calculator to find the tangent (tan) of each reference angle. Round to the nearest thousandth.

Angle Measure	Tan
10°	0.176
20°	0.364
30°	0.577
40°	0.839
50°	1.192
60°	1.732
70°	2.747
80°	5.671



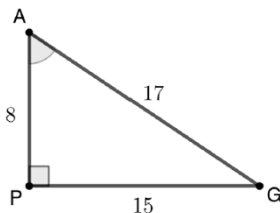
This Exploration is scaffolded to allow students to engage with their partner rather than being guided through by the teacher. There will be an opportunity to summarize and synthesize what is discovered in the next component. Students may revise their thinking throughout the exploration as they engage with each new reflection.

7.3.2 Exploration: Relating Sine, Cosine and Tangent (continued)

3. Write down one notice statement and one wonder statement about the values of $\frac{\sin}{\cos}$ from the first table and the values of $\tan$ from the second table.

Notice	Wonder
<i>Answers will vary. I notice that the tangent value is approximately the same as the sine divided by cosine for the same angle measurements</i>	<i>Answers will vary. Why does the sine divided by the cosine result in the tangent?</i>

4. $\triangle APG$ is shown, where $m\angle APG = 90^\circ$, $AP = 8$, $PG = 15$, and $AG = 17$.



- a. Using $\angle PAG$ as the reference angle, complete the table.

Sine $\left(\frac{\text{opposite}}{\text{hypotenuse}}\right)$	Cosine $\left(\frac{\text{adjacent}}{\text{hypotenuse}}\right)$	Tangent $\left(\frac{\text{opposite}}{\text{adjacent}}\right)$
$\frac{15}{17} \approx 0.882$	$\frac{8}{17} \approx 0.471$	$\frac{15}{8} = 1.875$

- b. Multiply the sine and the cosine ratios from the table.

$$\frac{15}{17} \cdot \frac{8}{17} = \frac{120}{289}$$



If students are finding that 10° is 0.648, then they have their calculators set in radian mode. Model for students how to change the calculator setting to degree.



Think-Pair-Share

Consider using this instructional routine for the Notice and Wonder. Allow students the opportunity to think about things they notice and wonder before sharing with a partner. A class discussion can ensure that students are noticing the relationship between the values of the ratio of sine and cosine and the values of tangent.



For a visual approach, allow students to color code the opposite leg, adjacent leg, and hypotenuse based on the reference angle. Using the same color coding, students can write the ratios for sine, cosine, and tangent.



Common Misconception

Students may get $\frac{120}{17}$ for the answer. Facilitate a discussion with the class about how to multiply fractions. If students are still struggling with fractions, consider allowing them to complete On-Ramp to Algebra 1: Multiplying Two Fractions.

7.3.2 Exploration: Relating Sine, Cosine and Tangent (continued)

- c. Rewrite $\text{sine} \cdot \text{cosine}$ using the ratios of sine and cosine, $\sin(\theta) = \frac{\text{opposite}}{\text{hypotenuse}}$ and $\cos(\theta) = \frac{\text{adjacent}}{\text{hypotenuse}}$. If possible, simplify.

$$\begin{aligned}\sin(\theta) \cdot \cos(\theta) &= \frac{\text{opposite}}{\text{hypotenuse}} \cdot \frac{\text{adjacent}}{\text{hypotenuse}} \\ &= \frac{\text{opposite}(\text{adjacent})}{\text{hypotenuse}^2}\end{aligned}$$

- d. Divide the sine ratio by the cosine ratio from the table.

$$\sin(\theta) \div \cos(\theta) = \frac{15}{17} \div \frac{8}{17} = \frac{15}{8}$$

- e. Rewrite $\frac{\text{sine}}{\text{cosine}}$ using the ratios of sine and cosine, $\sin(\theta) = \frac{\text{opposite}}{\text{hypotenuse}}$ and $\cos(\theta) = \frac{\text{adjacent}}{\text{hypotenuse}}$. The resulting ratio is a complex fraction that can be rewritten or simplified. Work with your partner to write the simplified form.

$$\frac{\sin(\theta)}{\cos(\theta)} = \frac{\frac{\text{opposite}}{\text{hypotenuse}}}{\frac{\text{adjacent}}{\text{hypotenuse}}} = \frac{\text{opposite}}{\text{hypotenuse}} \cdot \frac{\text{hypotenuse}}{\text{adjacent}} = \frac{\text{opposite}}{\text{adjacent}}$$

- f. Describe how the ratios of sine and cosine can be used to find the tangent ratio.

Answers may vary. In order to find the tangent ratio of any right triangle, divide the ratio of sine by the ratio of cosine of the same reference angle.



For a kinesthetic approach, give students a set of 15 index cards with the following written on them:

- 5 - opposite
- 5 - adjacent
- 5 - hypotenuse

Students can use the cards to look at how to multiply and divide fractions instead of writing it out.



After question 4e, consider asking the following.

- What happened to the hypotenuse when dividing sine by cosine?
- If you are only given sine and cosine, can you find tangent? If so, how?
- If you are only given cosine and tangent, can you find the sine? If so, how?

7.3.3 Guided Instruction: Determining Trigonometric Ratios

Component Overview: Students are finding and comparing the sine, cosine, and tangent from given reference angles.

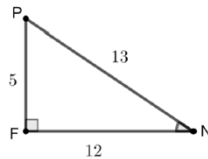


7.3.3 Guided Instruction: Determining Trigonometric Ratios



The tangent ratio of a reference angle can be found using the ratio, $\frac{\text{sine}}{\text{cosine}}$.

1. $\triangle PFN$ is shown, where $m\angle PFN = 90^\circ$, $PF = 5$, $FN = 12$, and $PN = 13$.



- a. Using $\angle FNP$ as the reference angle, find the three ratios.

Cosine	Sine	Tangent
$\frac{12}{13}$	$\frac{5}{13}$	Answers may vary. $\frac{\left(\frac{5}{13}\right)}{\left(\frac{12}{13}\right)} = \frac{5}{13} \cdot \frac{13}{12} = \frac{5}{12}$ Or: $\frac{5}{12}$

- b. Which method, $\frac{\text{sine}}{\text{cosine}}$ or $\frac{\text{opposite}}{\text{adjacent}}$, did you use to find the tangent ratio?

Answers may vary.

$$\frac{\text{sine}}{\text{cosine}} = \frac{\frac{5}{13}}{\frac{12}{13}} = \frac{5}{13} \cdot \frac{13}{12} = \frac{5}{12}$$

Or:

$$\frac{\text{opposite}}{\text{adjacent}} = \frac{5}{12}$$



If tangent is the ratio of sine to cosine, what sides of a right triangle can also be used to write a ratio for the tangent?



Common Misconception

Check to ensure students have correctly identified sine and cosine. If students have written the reciprocal of the ratios, then they may have mislabeled the sides. Consider having students color code the sides and/or draw an arrow from the reference angle to the side across from it in order to identify the opposite side.

7.3.3 Guided Instruction: Determining Trig Ratios (continued)

- c. Use the method for finding tangent that you did not use to check the ratio.

Answers may vary.

$$\frac{\text{sine}}{\text{cosine}} = \frac{\frac{5}{13}}{\frac{12}{13}} = \frac{5}{13} \cdot \frac{13}{12} = \frac{5}{12}$$

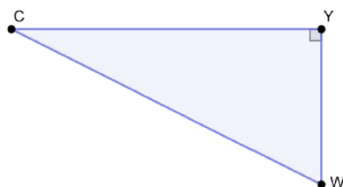
Or:

$$\frac{\text{opposite}}{\text{adjacent}} = \frac{5}{12}$$



Monitor student responses and consider having 2-3 students who used different methods share their responses.

2. $\triangle CYW$ is shown, where $CY = \sqrt{98}$ and $YW = \sqrt{28}$.



- a. Determine the length of $\overline{CW}$.

$$\sqrt{98}^2 + \sqrt{28}^2 = (CW)^2$$

$$CW = \sqrt{126} = 3\sqrt{14}$$

- b. Using $\angle WCY$ as the reference angle, find the three ratios.

Cosine	Sine	Tangent
$\frac{\sqrt{98}}{\sqrt{126}}$	$\frac{\sqrt{28}}{\sqrt{126}}$	$\frac{\sqrt{28}}{\sqrt{98}}$

- c. Using $\angle YWC$ as the reference angle, find the three ratios.

Cosine	Sine	Tangent
$\frac{\sqrt{28}}{\sqrt{126}}$	$\frac{\sqrt{98}}{\sqrt{126}}$	$\frac{\sqrt{98}}{\sqrt{28}}$

7.3.3 Guided Instruction: Determining Trig Ratios (continued)

- d. What do you notice about the ratios for $\angle WCY$ and the ratios for $\angle YWC$?

Answers will vary.

The sine of one angle is the cosine of the other reference angle.

The tangent ratios are reciprocals. The numerator for the sine and cosine swap when swapping the reference angles.



Have students share with their partner or whole group what they notice about the ratios.

7.3.4 Guided Practice: Trigonometric Ratios

Component Overview: After using the Pythagorean Theorem to find missing sides of right triangles, students will write the ratios of sine, cosine, and tangent for a given reference angle.



7.3.4 Guided Practice: Trigonometric Ratios

1. $\triangle YNB$, $m\angle NBY = 90^\circ$, $BN = 6\sqrt{2}$, and $YN = 4\sqrt{17}$.

- a. Determine the length of $\overline{YB}$.

$$(6\sqrt{2})^2 + (YB)^2 = (4\sqrt{17})^2$$

$$YB = \sqrt{200} = 10\sqrt{2}$$

- b. Using $\angle BNY$ as the reference angle, find the three ratios.

Cosine	Sine	Tangent
$\frac{6\sqrt{2}}{4\sqrt{17}} = \frac{3\sqrt{2}}{2\sqrt{17}}$	$\frac{10\sqrt{2}}{4\sqrt{17}} = \frac{5\sqrt{2}}{2\sqrt{17}}$	$\frac{10\sqrt{2}}{6\sqrt{2}} = \frac{5}{3}$

2. $\triangle GMK$, $m\angle GKM = 90^\circ$, $GM = 3\sqrt{10}$, and $GK = 3\sqrt{3}$.

- a. Determine the length of $\overline{KM}$.

$$(3\sqrt{3})^2 + (KM)^2 = (3\sqrt{10})^2$$

$$KM = \sqrt{63} = 3\sqrt{7}$$

- b. Using $\angle KGM$ as the reference angle, find the three ratios.

Cosine	Sine	Tangent
$\frac{3\sqrt{3}}{3\sqrt{10}} = \frac{\sqrt{3}}{\sqrt{10}}$	$\frac{3\sqrt{7}}{3\sqrt{10}} = \frac{\sqrt{7}}{\sqrt{10}}$	$\frac{3\sqrt{7}}{3\sqrt{3}} = \frac{\sqrt{7}}{\sqrt{3}}$



Provide students with copies of right triangles with no sides or angles labeled. Task students with appropriately labeling the triangle prior to answering the questions.



Common Misconception

If students find the missing side to be $\sqrt{3}$, then students did not square 3 when squaring $3\sqrt{10}$. Facilitate a discussion about how to square radicals when there is a coefficient.

7.3.5 Wrap-Up: Cool Down

Component Overview: Students write a short summary of their learning in the lesson. Student responses can be used to identify misconceptions and plan for upcoming instruction.



7.3.5 Wrap-Up: Cool Down

1. Write a short note to a friend who missed today's class summarizing what you learned.

Answers will vary.



The information provided here, coupled with what was observed during instruction, can provide information for differentiating instruction in later lessons.

7.4 Trigonometric Ratios on the Coordinate Plane

Lesson Introduction

 Benchmarks	MA.912.T.1.1 Define trigonometric ratios for acute angles in right triangles. MA.912.T.1.2 Solve mathematical and real-world problems involving right triangles using trigonometric ratios and the Pythagorean Theorem.		 Learning Target	I can determine trigonometric ratios using the coordinate plane.
 Benchmark Coherence	Building On MA.8.GR.1.1 Apply the Pythagorean Theorem to solve mathematical and real-world problems involving unknown side lengths in right triangles.		Working Toward N/A	
 Essential Questions	- How are trigonometric ratios related in similar triangles? - How do I find trigonometric ratios using a coordinate plane?			
 Lesson Narrative	Through work on the coordinate plane, students discover the relationship between trigonometric ratios in similar triangles. Using the knowledge that trigonometric ratios are equivalent in similar triangles, students will find trigonometric ratios of given reference angles.			
 Component Summary	7.4.1 Warm-Up (~5 min.)	Describe a pattern (<i>SE p. 388</i>)	7.4.3 Guided Instruction (10-15 min.)	Approximate trigonometric ratios (<i>SE p. 390</i>)
	7.4.2 Collaboration (10-15 min.)	Define trigonometric ratios on the coordinate plane (<i>SE p. 388</i>)	7.4.4 Guided Practice (5 min.)	Find trigonometric ratios (<i>SE p. 392</i>)
	7.4.5 Wrap-Up (~5 min.)	Explain the relationship between trigonometric ratios and similar triangles (<i>SE p. 393</i>)		
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> - Sine - Cosine - Tangent		- Scientific calculator (optional) Colored Pencils (optional) Highlighters	N/A

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may flip the answers for sine and cosine if they have mislabeled the sides of a right triangle. - Students may forget to take the square root when using the Pythagorean Theorem to solve. - Students may divide by 2 when using the Pythagorean Theorem to solve, instead of taking the square root.	Consider allowing students to trace triangles on patty paper if they are struggling with solving triangles that are overlapping.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Think-Pair-Share After completing the chart with the trigonometric ratios in 7.4.2 Collaboration, have the students pause to allow them to ponder the question, “What similarities do they notice?” Students may then work with a partner to answer the question and share with the class or surrounding groups.	MA.K12.MTR.2.1: Multiple Representations Throughout the lesson, spotlight the different ways trigonometric ratios can be represented through similar triangles. Consider using the language embedded in the benchmark clarifications for MA.K12.MTR.2.1 when facilitating discussion.
 Differentiate Instruction	Using color coding methods when working with similar triangles will help visual students keep their work organized and draw conclusions from their findings in the collaboration component of this lesson. The 7.4.2 Collaboration allows entry points for auditory learners to discuss and share their findings with a partner, small group, or the class as a whole. Additional differentiation opportunities are denoted throughout the lesson guide.	
Lesson Notes:		

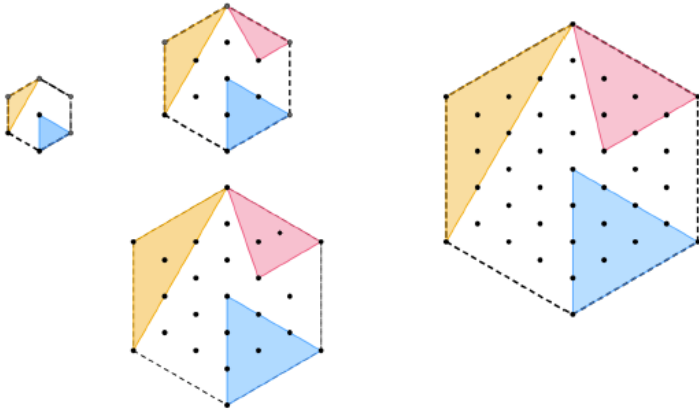
7.4.1 Warm-Up: Patterns

Component Overview: Students are observing and stating the patterns seen in the images given.



7.4.1 Warm-Up: Patterns

Four hexagons are shown.



For a different entry point for students who struggle writing, provide the next two blank hexagons with the dots for students to draw the 5th and 6th patterns. Then have students verbally state how they knew to draw the patterns.

1. Describe the pattern of each triangle.

Triangle	Pattern Description
Yellow	<i>The triangle appears to be isosceles. In each shape in the sequence, the sides seem to be increasing by 1 unit dot each time.</i>
Red	<i>The triangle appears to be scalene. In each shape in the sequence, the sides seem to be increasing by 1 unit dot each time.</i>
Blue	<i>The triangle appears to be equilateral. In each shape in the sequence, the sides seem to be increasing by 1 unit dot each time.</i>

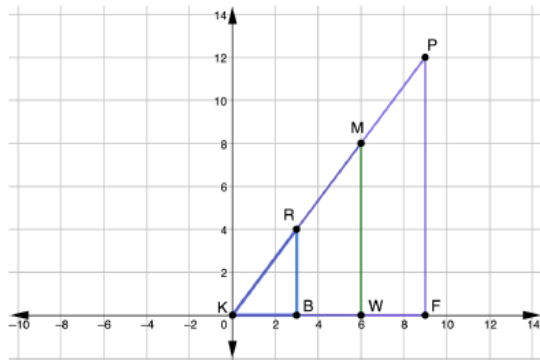
7.4.2 Collaboration: Defining Trigonometric Ratios

Component Overview: Through collaboration with peers, students are working together to discover the relationship between trigonometric ratios and similar triangles on the coordinate plane.



7.4.2 Collaboration: Defining Trigonometric Ratios

- Three right triangles are shown on the coordinate grid.



- Discuss with your partner how to justify that ΔKBR , ΔKWM , and ΔKFP are similar. Summarize your discussion.

Answers will vary. The three triangles are similar by AA similarity because they share $\angle K$ and they all have a right angle. Also, the sides are proportional.

- Complete the table.

Segment	Length	Segment	Length
$\overline{KB}$	$KB = 3$	$\overline{WM}$	$WM = 8$
$\overline{BR}$	$BR = 4$	$\overline{KF}$	$KF = 9$
$\overline{KW}$	$KW = 6$	$\overline{FP}$	$FP = 12$



Consider selecting students who used different methods and proved similarity correctly to discuss why each method is correct. If all students used the same method, consider asking students to prove similarity using a different method.

7.4.2 Collaboration: Defining Trigonometric Ratios (continued)

- c. Use the lengths from the table to determine each. Show your work.

Segment	Length
$\overline{KR}$	$3^2 + 4^2 = (KR)^2$ $9 + 16 = (KR)^2$ $25 = (KR)^2$ $\sqrt{25} = KR$ $5 = KR$
$\overline{KM}$	$6^2 + 8^2 = (KM)^2$ $36 + 64 = (KM)^2$ $100 = (KM)^2$ $\sqrt{100} = \sqrt{(KM)^2}$ $10 = KM$
$\overline{KP}$	$9^2 + 12^2 = (KP)^2$ $81 + 144 = (KP)^2$ $225 = c^2$ $\sqrt{225} = \sqrt{KP^2}$ $\sqrt{225} = KP$ $15 = KP$

- d. Using $\angle K$ as the reference angle, find the three ratios for each triangle.

Triangle	Cosine	Sine	Tangent
$\triangle RKB$	$\frac{3}{5}$	$\frac{4}{5}$	$\frac{4}{3}$
$\triangle MKW$	$\frac{6}{10}$	$\frac{8}{10}$	$\frac{8}{6}$
$\triangle PKF$	$\frac{9}{15}$	$\frac{12}{15}$	$\frac{12}{9}$



For a visual approach, have students highlight the triangle segments in different colors.



Common Misconception
Students may incorrectly solve using the Pythagorean Theorem by not taking the square root to find the hypotenuse.



Think-Pair-Share
After completing the chart with the trigonometric ratios, allow students to ponder the question, "What similarities do you notice with the ratios?" Students may then work with a partner to answer the question and share with the class or surrounding groups.

7.4.2 Collaboration: Defining Trigonometric Ratios (continued)

2. $\triangle HWB$ is graphed on the coordinate grid, where $m\angle WBH = 90^\circ$. The coordinates of the vertices are $H(-4, 5)$, $W(5, -3)$, and $B(-4, -3)$.
- a. Discuss with your partner how to use the coordinates to determine the trigonometric ratios. Summarize your discussion.
Answers will vary.
I can use the distance between the points to determine the length of the sides. Then, using these lengths, I can determine the trigonometric ratios.
- b. Complete the table. Show your work.

Segment	Length
$\overline{WB}$	$WB = 9$
$\overline{BH}$	$BH = 8$
$\overline{HW}$	$HW = \sqrt{145}$

- c. Using $\angle WHB$ as the reference angle, find the three ratios.

Cosine	Sine	Tangent
$\frac{8}{\sqrt{145}}$	$\frac{9}{\sqrt{145}}$	$\frac{9}{8}$



Consider providing struggling students graph paper to use as a visual to help them conceptualize using coordinates to determine the trig ratios.



Consider determining how you would like students to answer the distance of HW .



Common Misconception

If students find that $HW = 72.5$, then students divided by 2 instead of taking the square root. Consider facilitating a discussion with the class or a small group of students about how powers and radicals are inverse operations.

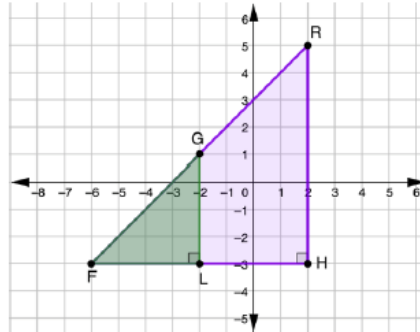
7.4.3 Guided Instruction: Using Trigonometric Ratios

Component Overview: The guided instruction works through how to find trigonometric ratios and how to approximate the value of the ratio.



7.4.3 Guided Instruction: Using Trigonometric Ratios

1. $\triangle FLG$ and $\triangle FHR$ shown on the coordinate grid are isosceles right triangles.



- a. Determine the measure of each angle.

Angle	Measure
$\angle LFG$	45°
$\angle FGL$	45°
$\angle FRH$	45°

- b. Using $\angle F$ as the reference angle, find the three ratios for each triangle.

Triangle	Cosine	Sine	Tangent
$\triangle FLG$	$\frac{4}{\sqrt{32}}$	$\frac{4}{\sqrt{32}}$	$\frac{4}{4} = 1$
$\triangle FHR$	$\frac{8}{\sqrt{128}}$	$\frac{8}{\sqrt{128}}$	$\frac{8}{8} = 1$

- c. Use your calculator to find each. Round to the nearest thousandth, if necessary.

Angle	Cosine	Sine	Tangent
45°	0.707	0.707	1

- d. How are the ratios you found using $\angle F$ as the reference angle related to the values you found for the ratios of a 45° angle?

The values found using the calculator for a 45° angle are equivalent to the decimal approximations of the ratios found using $\angle F$ as a reference angle.

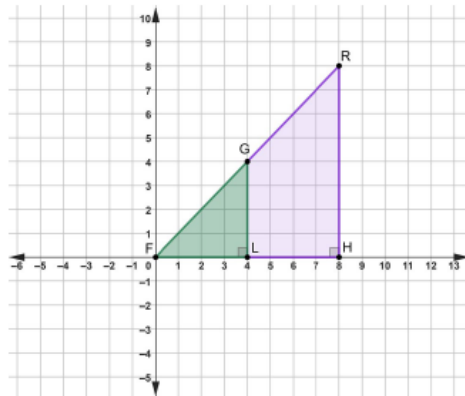


- What does it mean for a triangle to be an isosceles triangle?
- How can you find the base angles in a right isosceles triangle?

7.4.3 Guided Instruction: Using Trigonometric Ratios (continued)

- e. $\triangle FLG$ and $\triangle FHR$ were moved on the coordinate grid so that F is at the origin.

Work with your partner to determine whether or not the values of the trigonometric ratios will change because the triangle moved. Summarize your findings.
The values will not change, for the figure was only repositioned by a rigid transformation and not resized.



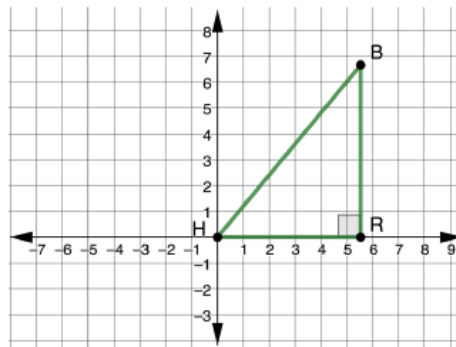
Common Misconception

If students find cosine and sine are 1.41, then students used the reciprocal of the trigonometric ratios. This misconception can be addressed by discussing the ratios for each trigonometric function, as well as how to correctly label a right triangle based on a reference angle.



When using a calculator, consider reminding students to make sure their calculator is in degrees, not radians.

2. $\triangle HBR$ is shown on the coordinate grid, where
 $m\angle HRB = 90^\circ$,
 $m\angle RHB = 50^\circ$,
 $HR = 5.567$, and
 $HB = 8.66$.



- a. Explain why the ratio $\cos(50^\circ) \approx \frac{5.567}{8.66}$ is true.

Answers may vary.

In a right triangle, the cosine of an angle is the ratio of $\frac{\text{adjacent}}{\text{hypotenuse}}$. In $\triangle HBR$, the length of the side adjacent to $\angle RHB$, which has the measure of 50° , is 5.567 and the hypotenuse of the triangle is $\overline{HB}$, which has a length of 8.66.

7.4.3 Guided Instruction: Using Trigonometric Ratios (continued)

- b. Find each value using a calculator. Round to the nearest thousandth.

$$\cos(50^\circ) = 0.643 \qquad \frac{5.567}{8.66} = 0.643$$

- c. Discuss with your partner why the approximation symbol was used in $\cos(50^\circ) \approx \frac{5.567}{8.66}$. Summarize your thoughts below.

The value of both $\cos 50^\circ$ and $\frac{5.567}{8.66}$ are approximately the same. Using a calculator to approximate each shows that $\cos 50^\circ$ is 0.6427876097 and $\frac{5.567}{8.66}$ is 0.6428406467. These two values are approximately the same and both have the same value if rounded to the thousandths.

- d. Work with your partner to find $\sin(H)$. Round to the nearest thousandth.

$$5.567^2 + (RB)^2 = 8.66^2$$

$$RB \approx 6.634$$

$$\sin(H^\circ) = \frac{6.634}{8.66}$$

$$\sin(H^\circ) \approx 0.766$$

- e. Use your calculator to find $\sin(50^\circ)$.

$$\sin(50^\circ) \approx 0.766$$

- f. Explain why $\sin(50^\circ) \approx \sin(H)$.

Both of these values when rounded equal approximately 0.766 so they are approximately equal to each other.

- g. Work with your partner to write the coordinates of each vertex.

Vertex	Coordinates
R	(5.567, 0)
B	(5.567, 6.634)



Monitor students as they discuss with their partner, and identify pairs of partners whom you wish to share with the class.

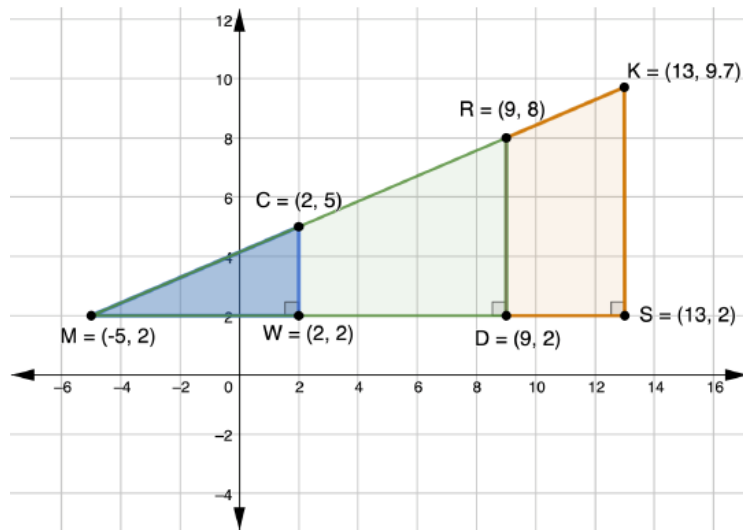
7.4.4 Guided Practice: Trigonometric Ratios and Similar Triangles

Component Overview: Through this teacher-led practice, students will practice approximating trigonometric ratios.



7.4.4 Guided Practice: Trigonometric Ratios and Similar Triangles

1. $\triangle CMW$, $\triangle RMD$, and $\triangle KMS$ are shown on the coordinate grid.



Using $\angle M$ as the reference angle, find the three ratios for each triangle. Write the values as a decimal rounded to the nearest thousandth.

Triangle	Cosine	Sine	Tangent
$\triangle CMW$	0.919	0.394	0.429
$\triangle RMD$	0.919	0.394	0.429
$\triangle KMS$	0.919	0.394	0.429

2. Yasmeen said she only found the ratios for $\triangle CMW$ then applied them to the other triangles. Explain using mathematics why she can do that.

All three triangles are similar by AA similarity therefore all of their sides are proportional.



Allow informal observation throughout the first two activities to dictate how “guided” this guided practice should be for you and your classroom. This could mean small group instruction for some while others attempt independently, or it could mean teacher facilitation of practice with students actively working as a whole group.



For a kinesthetic approach, consider allowing students to copy the three triangles on to separate graphs. Students can label the different side lengths accordingly.



Common Misconception

Monitor student answers to check that sine and cosine are not being switched. If they are switched, it may indicate students have mislabeled the sides. Consider facilitating a discussion about how to label the sides of the triangle as opposite, adjacent, and hypotenuse.

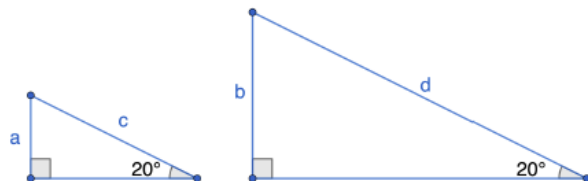
7.4.5 Wrap-Up: Ticket Out the Door

Component Overview: Students will complete a Ticket Out the Door that demonstrates their understanding of the relationship between trigonometric ratios and similar triangles.



7.4.5 Wrap-Up: Ticket Out the Door

1. Two triangles are shown.



Use student responses along with teacher observations to plan for remediation and future lessons.

- a. Describe how the ratios $\frac{a}{c}$ and $\frac{b}{d}$ are related.

$\frac{a}{c} = \frac{b}{d}$ because the triangles are similar and therefore their sides are proportional.

- b. What mathematical relationship(s) did you use to complete your description?

Answers may vary. Since the triangles are similar by the AA theorem, the ratios will simplify to equivalent values.

7.5 Using Trigonometric Ratios

Lesson Introduction

Lesson Introduction				
 Benchmark	MA.912.T.1.2 Solve mathematical and real-world problems involving right triangles using trigonometric ratios and the Pythagorean Theorem.		 Learning Targets	- I can use trigonometric ratios to find the side of a right triangle. - I can use trigonometric ratios to find an angle of a right triangle.
 Benchmark Coherence	Building On: MA.8.GR.1.1 Apply the Pythagorean Theorem to solve mathematical and real-world problems involving unknown side lengths in right triangles.		Working Toward: N/A	
 Essential Questions	- How can I use trigonometric ratios to find the side of a right triangle? - How can I use trigonometric ratios to find an angle of a right triangle?			
 Lesson Narrative	In this lesson, students build on the previous lessons by using trigonometric ratios to determine the value of missing sides. They will then be introduced to using inverse trigonometric functions to find the missing angle measure in a right triangle and how to use trigonometric ratios to find missing side measures of a right triangle.			
 Component Summary	7.5.1 Warm-Up (~5 min.)	Activate prior knowledge (<i>SE p. 394</i>)	7.5.4 Guided Instruction (10 min.)	Use trigonometric ratios to find measures (<i>SE p. 397</i>)
	7.5.2 Collaboration (10 min.)	Use trigonometric ratios to write equations (<i>SE p. 394</i>)	7.5.5 Guided Practice (5 min.)	Use trigonometric ratios to find measures (<i>SE p. 398</i>)
	7.5.3 Guided Instruction (10-15 min.)	Use inverse trigonometric functions to find angle measures (<i>SE p. 396</i>)	7.5.6 Wrap-Up (~5 min.)	Find the measure of a missing side and a missing angle (<i>SE p. 399</i>)
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> - Inverse Trigonometric Functions - Arccosine - Arcsine - Arctangent <u>Additional Terms:</u> N/A		Scientific Calculator	N/A

 Teacher Tips	Common Misconceptions - Students may use trigonometric functions to find missing angle measures instead of inverse trigonometric functions. - Students may incorrectly use their calculator with trigonometric function keys when completing calculations.	Things to Consider Students have learned about trigonometric ratios in the prior lesson. In this lesson, students will be using them to solve problems. Some students may need remediation on the basics of trigonometric functions.
	Literacy and Language Routines Compare and Connect This lesson consists of two unique activities that are teacher led, 7.5.3 and 7.5.4 Guided Instructions. During this process, the opportunity for teachers to share their thinking out loud will be available. Students will be prompted to reflect and respond. This routine supports meta-cognitive and meta-linguistic awareness and also supports mathematical conversation.	Mathematical Thinking and Reasoning (MTR) Standard MA.K12.MTR.5.1: Patterns and Structure Throughout this lesson, students will use patterns and structures to help understand and connect mathematical concepts. Students will decompose complex problems into manageable parts and relate previously learned concepts in earlier lessons to new concepts in this lesson.
	 Differentiate Instruction	
Consider the creation of a foldable that provides a difference between the use of trigonometric functions to solve a problem (one for each function) and using inverse functions to solve problems (one for each function), so students can have a reference guide for the two scenarios. Additional differentiation opportunities are denoted throughout the lesson guide.		
Lesson Notes:		

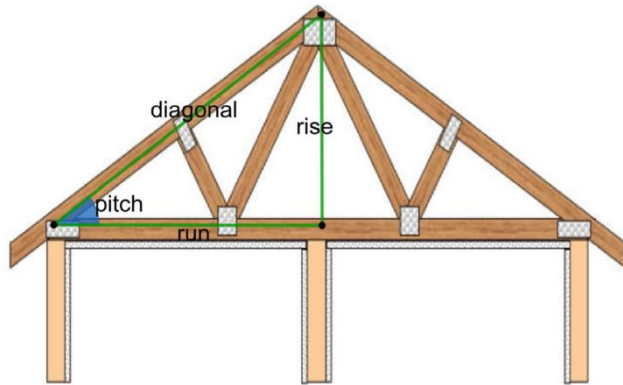
7.5.1 Warm-Up: Notice and Wonder

Component Overview: Students activate prior knowledge of right triangles through a Notice and Wonder.



7.5.1 Warm-Up: Notice and Wonder

In carpentry, right triangles are commonly used in constructing roofs. A roof's pitch is the steepness of the roof. It can be determined by the ratio between the two legs of the right triangle.



Prompt students to make the connection between this question and the Pythagorean Theorem, which we have been working on in the last few lessons.

1. What do you notice or wonder about the use of right triangles in construction?

Answers will vary. I notice that the triangles used in the picture are equilateral and are right triangles. I wonder what other types of triangles are commonly used in carpentry.

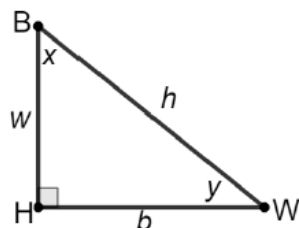
7.5.2 Collaboration: Using Trigonometric Ratios

Component Overview: Students will work together to write equations using trigonometric ratios.



7.5.2 Collaboration: Using Trigonometric Ratios

1. $\triangle BWH$ is shown, where
 $h = 12.8$, $w = 8$,
 $x = 51.3^\circ$, and $y = 38.7^\circ$.



- a. Select all of the equations that can be used to find the value of b .

☐ $\cos(51.3^\circ) = \frac{8}{b}$ ☐ $\sin(51.3^\circ) = \frac{b}{12.8}$ ☐ $\tan(51.3^\circ) = \frac{b}{12.8}$

☐ $\cos(38.7^\circ) = \frac{b}{12.8}$ ☐ $\sin(38.7^\circ) = \frac{b}{12.8}$ ☐ $\tan(38.7^\circ) = \frac{8}{b}$

- b. Determine the value of b . Round to the nearest thousandth.

$$\sin(51.3^\circ) = \frac{b}{12.8}$$

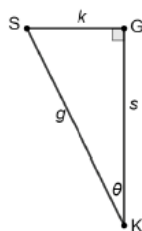
$$b = 9.990$$

2. $\triangle SGK$ is shown.

- a. Use a trigonometric ratio for θ to write an equation that can be used to find the value of k .

$$\sin(\theta) = \frac{k}{g}$$

$$\tan(\theta) = \frac{k}{s}$$



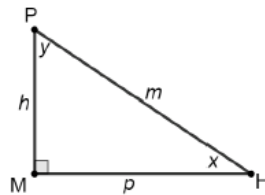
Student responses to question 1a can provide formative data to determine which students may need additional support to remediate the information learned in the prior lessons in this unit.

7.5.2 Collaboration: Using Trigonometric Ratios (continued)

- b. Which trigonometric ratio for θ cannot be used to write an equation that can be used to find the value of k ?

$\cos(\theta)$ since $\cos(\theta) = \frac{\text{adjacent}}{\text{hypotenuse}} = \frac{s}{g}$.

3. $\triangle PMH$ is shown. Work with your partner to determine which trigonometric ratio can be used to determine the unknown side or angle. Then, use the ratio to write an equation that can be used to determine the unknown. When writing the equation, use the variables h , p , m , x , and y .



Given	Unknown	Trigonometric Ratio	Equation
$m\angle PHM$ and length of $\overline{PM}$	$\overline{PH}$	☐ $\sin$ ☐ $\cos$ ☐ $\tan$	$\sin(x) = \frac{h}{m}$
$m\angle PHM$ and length of $\overline{PM}$	$\overline{MH}$	☐ $\sin$ ☐ $\cos$ ☐ $\tan$	$\tan(x) = \frac{h}{p}$
$m\angle HPM$ and length of $\overline{PM}$	$\overline{PH}$	☐ $\sin$ ☐ $\cos$ ☐ $\tan$	$\cos(y) = \frac{h}{m}$
Lengths of $\overline{MH}$ and $\overline{PH}$	$m\angle HPM$	☐ $\sin$ ☐ $\cos$ ☐ $\tan$	$\sin(y) = \frac{p}{m}$
Lengths of $\overline{MH}$ and $\overline{PM}$	$m\angle PHM$	☐ $\sin$ ☐ $\cos$ ☐ $\tan$	$\tan(x) = \frac{h}{p}$



How did you determine which trigonometric ratio could be used to determine the value of k ?

7.5.3 Guided Instruction: Using Inverse Trigonometric Functions to Find Angle Measure

Component Overview: Students will be introduced to using the inverse of trigonometric functions to find angle measures through three multi-part questions.



7.5.3 Guided Instruction: Using Inverse Trigonometric Functions to Find Angle Measure

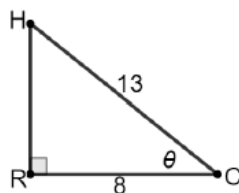


Inverse Trigonometric Functions

$\cos^{-1}\left(\frac{a}{c}\right)$ is arccosine. $\sin^{-1}\left(\frac{b}{c}\right)$ is arcsine. $\tan^{-1}\left(\frac{b}{a}\right)$ is arctangent. These functions can be used to determine the measure of an angle where a is the side adjacent the angle, b is the side opposite the angle, and c is the hypotenuse. The calculator should be in degree mode.

1. In $\triangle HRC$, $m\angle HRC = 90^\circ$, $RC = 8$, and $HC = 13$.

The equation $\cos(\theta) = \frac{8}{13}$ can be used to determine θ . To find θ using a calculator use the $\cos^{-1}\left(\frac{8}{13}\right)$. Round any angle measurements below to the nearest whole degree.



- a. What is $\cos^{-1}\left(\frac{8}{13}\right)$?

$$\cos^{-1}\left(\frac{8}{13}\right) \approx 52^\circ$$

- b. Write an equation that can be used to find $m\angle CHR$.

$$m\angle CHR = \sin^{-1}\left(\frac{8}{13}\right)$$

- c. Find $m\angle CHR$ using an inverse trigonometric function on a calculator. Be sure the calculator is in degree mode.

$$m\angle CHR = \sin^{-1}\left(\frac{8}{13}\right) \approx 38^\circ$$



Consider the creation of a foldable that provides a difference between the use of trigonometric functions to solve a problem (one for each function) and using inverse functions to solve a problem (one for each function), so students can have a reference guide for the two scenarios.



Students may need additional support with inputting the inverse trig function into their calculator. Consider modeling with a digital calculator, such as Desmos scientific calculator, or using a document camera, to show students how to input the values into their calculators. If needed, consider giving them a few to try to ensure students all get the same answer.



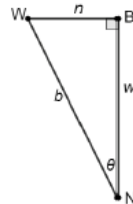
Why can this equation be used to determine the value of the unknown angle? Could another one be used? Why or why not?

7.5.3 Guided Instruction: Using Inverse Trigonometric Functions to Find Angle Measure (continued)

2. $\triangle WLT$ is a right triangle with $m\angle WTL = 90^\circ$, $TL = 11$, and $WL = 23$. What is the $m\angle TWL$? Round to the nearest whole degree.

$$m\angle TWL = \sin^{-1}\left(\frac{11}{23}\right) \approx 29^\circ$$

3. $\triangle WBN$ is shown, where $\angle B$ is a right angle.



- a. Determine the measure of $\angle BNW$ if $n = 36$ and $w = 77$.

$$\angle BNW = \tan^{-1}\left(\frac{36}{77}\right) \approx 25^\circ$$

- b. Gabriella used $\sin^{-1}\left(\frac{77}{85}\right)$ to find the measure of $\angle BNW$. Determine if $\sin^{-1}\left(\frac{77}{85}\right)$ results in the same degree measure as your answer.

No, it does not:

$$\sin^{-1}\left(\frac{77}{85}\right) \approx 65^\circ.$$

- c. Discuss with your partner whether or not Gabriella's method is valid. Write a justification of your decision.

Answers may vary.

Gabriella's method is not valid. This is because she set up her arcsine formula incorrectly. Sine (or arcsine) is always $\frac{\text{opposite}}{\text{hypotenuse}}$, however, she used $\frac{\text{adjacent}}{\text{hypotenuse}}$, which is the cosine. It is due to this that resulted in Gabriella getting the incorrect value for the angle measurement.



At the conclusion of this activity, engage students in a discussion around their responses and thinking for questions 3b and 3c. This provides an opportunity to uncover potential misconceptions and provides students the opportunity to articulate their thinking and learn from their peers.

7.5.4 Guided Instruction: Using Trigonometric Ratios to Find Side Measures

Component Overview: Students will use trigonometric ratios to find side measures through three, multi-part questions.



7.5.4 Guided Instruction: Using Trigonometric Ratios to Find Side Measures

1. In $\triangle KDR$, $m\angle DKR = 31.2^\circ$, $m\angle KDR = 90^\circ$, and $KD = 17$. What is the length of $\overline{KR}$?

The trigonometric ratio to find the approximate length of KR will be cosine, since $\cos(\theta) = \frac{\text{adjacent}}{\text{hypotenuse}}$.

$$\cos(31.2^\circ) = \frac{17}{KR}$$

$$KR \cdot \cos(31.2^\circ) = 17$$

$$KR = \frac{17}{\cos(31.2^\circ)}$$

$$KR \approx 19.875$$

2. $\triangle MAU$ is a right triangle with $MA = 20.7$ and $MU = 9.1$. If $m\angle MUA = 90^\circ$, what is the $m\angle AMU$?

The trigonometric ratio to find $m\angle AMU$ will be cosine and arcsine, since $\cos(\theta) = \frac{\text{adjacent}}{\text{hypotenuse}}$. $\cos(\theta) = \frac{9.1}{20.7}$, which means that $\cos^{-1}\left(\frac{9.1}{20.7}\right) = \theta = m\angle AMU$.

$$\cos^{-1}\left(\frac{9.1}{20.7}\right) \approx 63.921^\circ$$

3. $\triangle NKD$ is shown, where $\angle K$ is a right angle, $KD = 36$, and $m\angle D = 65^\circ$

- a. Find the length of $\overline{ND}$. Show your work.

$$\cos(65^\circ) = \frac{36}{ND}$$

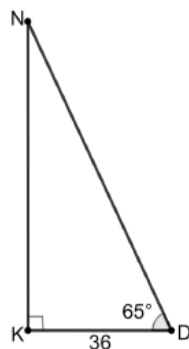
$$ND \cdot \cos(65^\circ) = \frac{36}{ND} \cdot ND$$

$$ND \cdot \cos(65^\circ) = 36$$

$$ND \cdot \frac{\cos(65^\circ)}{\cos(65^\circ)} = \frac{36}{\cos(65^\circ)}$$

$$ND = \frac{36}{\cos(65^\circ)}$$

$$ND \approx 85.183$$



This lesson consists of two unique guided instruction activities. During this process, the opportunity for teachers to share their thinking out loud will be available. Students will be prompted to reflect and respond. This routine supports meta-cognitive and meta-linguistic awareness and also supports mathematical conversation.



How did you determine which trigonometric function would help determine the angle in question 2?



How did you determine which trigonometric function would help determine the length of the side in question 3a?

7.5.4 Guided Instruction: Using Trigonometric Ratios to Find Side Measures (continued)

- b. Amelia and Henry found the length of $\overline{NK}$ using two different methods. The first step of their work is shown. Complete their work.

Amelia	Henry
$36^2 + NK^2 = (85.183)^2$	$\tan(65^\circ) = \frac{NK}{36}$
$1296 + NK^2 = 7256.1435$	
$1296 - 1296 + NK^2 = 7256.1435 - 1296$	$36 \cdot \tan(65^\circ) = \frac{NK}{36} \cdot 36$
$NK^2 = 5960.1435$	$36 \cdot \tan(65^\circ) = NK$
$\sqrt{NK^2} = \sqrt{5960.1435}$	
$NK \approx 77.20197$	$NK \approx 77.20225$

- c. Describe some possible reasons that Amelia's and Henry's answers are not exactly the same value.
Answers will vary.
Amelia used a rounded value for the hypotenuse.
- d. Whose method gives a precise answer?
Henry's answer is more precise.



Before moving to the next component, address any misunderstandings or clarifications that arise during 7.5.4 Guided Instruction.

7.5.5 Guided Practice: Using Trigonometric Ratios

Component Overview: Students will practice using trigonometric ratios to find missing angles and sides using two questions.



7.5.5 Guided Practice: Using Trigonometric Ratios

1. $\triangle ZME$ has $ME = 14.9$ and $m\angle ZEM = 41.5^\circ$.

- a. If $\angle ZME$ is a right angle, what is the length of $\overline{ZM}$?

$$\begin{aligned}\tan(41.5^\circ) &= \frac{ZM}{14.9} \\ 14.9 \cdot \tan(41.5^\circ) &= ZM\end{aligned}$$

$$ZM \approx 13.182$$

- b. Sharif used the information to determine the length of $\overline{ZM}$.
Her work is shown below.

$$\begin{aligned}\tan 41.5^\circ &= \frac{14.9}{ZM} \\ ZM \cdot \tan 41.5^\circ &= 14.9 \\ ZM &= \frac{14.9}{\tan 41.5^\circ} \\ ZM &\approx 16.841\end{aligned}$$

Find Sharif's error and correct it.

Sharif wrote the ratio of tangent as $\frac{\text{adjacent}}{\text{opposite}}$ instead of $\frac{\text{opposite}}{\text{adjacent}}$.

$$\begin{aligned}\tan 41.5^\circ &= \frac{ZM}{14.9} \\ 14.9 \cdot \tan 41.5^\circ &= ZM \\ ZM &\approx 13.182\end{aligned}$$

2. $\triangle KNP$ is a right triangle with $KP = 46$ and $KN = 27.3$. If $\angle PNK$ is a right angle, what is the $m\angle KPN$?
Since the opposite side length and the hypotenuse are given, arcsine can be used to find the angle measure.

$$m\angle KPN = \sin^{-1}\left(\frac{27.3}{46}\right) \approx 36.404^\circ$$



This activity provides the only practice opportunity with this material for this concept. Consider extending the practice opportunity or using one question as an independent practice opportunity prior to the Ticket Out the Door in the next activity.



- Why is the statement "is a right angle" critical to solving this problem?
- Could you still use this method if it was not? Why or why not?

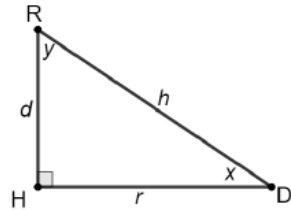
7.5.6 Wrap-Up: Ticket Out the Door

Component Overview: Students complete a Ticket Out the Door that has students find a missing angle and a missing side using trigonometric functions. Students should be allowed to access a calculator to complete the wrap-up.



7.5.6 Wrap-Up: Ticket Out the Door

1. $\triangle RDH$ is shown, where $\angle RHD$ is a right angle.



- a. If $d = 2.8$ and $h = 4.6$, what is the measure of y , in degrees?

$$y = \cos^{-1}\left(\frac{2.8}{4.6}\right) \approx 52.505^\circ$$

- b. If $x = 29^\circ$ and $h = 14.5$ what is the value of d ?

$$\begin{aligned}\sin(29^\circ) &= \frac{d}{14.5} \\ 14.5 \cdot \sin(29^\circ) &= d \\ d &\approx 7.03\end{aligned}$$



Student responses to this activity can provide data to which components of the lesson, if any, may need additional support and help to distinguish what students need additional assistance with understanding the content.

7.6 Special Right Triangles

Lesson Introduction

 Benchmark	MA.912.T.1.2 Solve mathematical and real-world problems involving right triangles using trigonometric ratios and the Pythagorean Theorem.		 Learning Targets	- I can use side ratios of 45–45–90 triangles to solve problems. - I can use side ratios of 30–60–90 triangles to solve problems.
 Benchmark Coherence	Building On MA.8.GR.1.1 Apply the Pythagorean Theorem to solve mathematical and real-world problems involving unknown side lengths in right triangles.		Working Toward	
 Essential Questions	- How can I use the side ratios of 45-45-90 triangles to solve problems? - How can I use the side ratios of 30-60-90 triangles to solve problems?			
 Lesson Narrative	In this lesson, students are introduced to the 45-45-90 and 30-60-90 special right triangles. Through exploration, students will discover the side relationships in special right triangles. They will then be formally introduced to 45-45-90 and 30-60-90 triangle formulas to find side lengths. Students will use this knowledge to solve problems involving these special right triangles.			
 Component Summary	7.6.1 Warm-Up (~5 min.)	Determine the shaded amount (<i>SE p. 400</i>)	7.6.3 Guided Instruction (5-10 min.)	Formalize the 45-45-90 and 30-60-90 triangle side relationships (<i>SE p. 405</i>)
	7.6.2 Exploration (20-25 min.)	Discover the side relationships in special right triangles (<i>SE p. 401</i>)	7.6.4 Guided Practice (5 min.)	Determine side measures in special right triangles (<i>SE p. 406</i>)
	7.6.5 Wrap-Up (~5 min.)	Demonstrate understanding of using special right triangles (<i>SE p. 407</i>)		
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> N/A		N/A	N/A

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may not recognize when a triangle meets the conditions of a special right triangle. - Students may apply the side relationships to the wrong special right triangles or other triangles.	- This lesson has fewer activities than some of the other lessons in this unit. Allow students ample time to engage in the 7.6.2 Exploration for students to discover the special triangles without giving too much up-front information. Students will see the formalized relationships in 7.6.3 Guided Instruction. - If time permits at the conclusion of the lesson, consider showing students a side-by-side comparison of using the side relationships in special right triangles and solving when the relationships are not used.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Think-Pair-Share 7.6.2 Exploration provides students with the opportunity to discover special right triangles. During this activity, students will have quiet time to think about the problems and work on them individually, and then time to share their response or their progress with a partner, and later the whole group.	MA.K12.MTR.6.1: Reasonableness This lesson requires students to perform multiple steps to find missing sides of special right triangles. Due to the problems requiring students to use order of operations, students may make minor errors that result in an incorrect solution. Encourage students to always check each step of their work and ask students to verify their solutions by explaining their work.
 Differentiate Instruction	Consider creating a side-by-side visual for determining side lengths using the side relationships of special right triangles versus not using the special right triangle side relationships, and the process for solving problems in both scenarios. Additional differentiation opportunities are denoted throughout the lesson guide.	
Lesson Notes:		

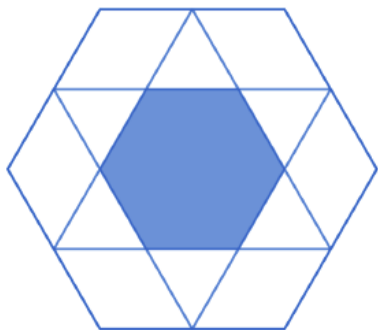
7.6.1 Warm-Up: Critical Thinking

Component Overview: Students will review the shading in a figure composed of multiple shapes in order to determine the fraction of the shape that is shaded.



7.6.1 Warm-Up: Critical Thinking

1. What fraction of the shape is shaded? $\frac{1}{4}$



2. Describe your problem-solving method.

Answers will vary.

Decompose the inner blue hexagon into congruent 6 congruent triangles. Then, decompose each parallelogram on the outer edges into 2 congruent triangles (6 parallelograms $\times$ 2 triangles). Count the total number of triangles that are in the entire large hexagon (24 triangles). 6 out of 24 triangles are $\frac{1}{4}$ of the hexagon.



Consider leveraging a Think-Pair-Share instructional routine with this activity. Allow students individual processing time to think through the problem on their own, and then work with a partner to discuss their answers and thinking around it to come to a consensus. Then have students share their thinking with the whole group.

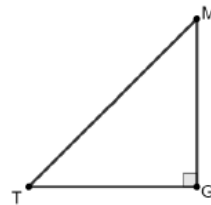
7.6.2 Exploration: Exploring Special Right Triangles

Component Overview: Students will explore 45-45-90 and 30-60-90 special right triangles to discover the side relationships.



7.6.2 Exploration: Exploring Special Right Triangles

1. $\triangle MGT$ is a right triangle, where $m\angle MGT = 90^\circ$, $MG = 6$, and $GT = 6$.



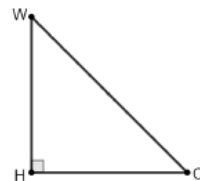
- a. What type of right triangle is $\triangle MGT$?

An isosceles right triangle

- b. Determine the measurements. Leave the measurement of $\overline{MT}$ in radical form.

MT	$m\angle GMT$	$m\angle GTM$
$(MG)^2 + (GT)^2 = (MT)^2$ $6^2 + 6^2 = (MT)^2$ $MT = 6\sqrt{2}$	$\tan^{-1}\left(\frac{6}{6}\right)$ 45°	$\sin^{-1}\left(\frac{6}{6\sqrt{2}}\right)$ 45°

2. $\triangle WCH$ is a right triangle, where $m\angle WHC = 90^\circ$, $m\angle WCH = 45^\circ$, and $WC = 4\sqrt{2}$.



- a. What type of right triangle is $\triangle WCH$?

An isosceles right triangle

- b. Determine the measurements of each.

HC	WH	$m\angle CWH$
$\cos(45^\circ) = \frac{HC}{4\sqrt{2}}$ $HC = 4$	$\sin(45^\circ) = \frac{WH}{4\sqrt{2}}$ $WH = 4$	$m\angle CWH + 45^\circ + 90^\circ = 180^\circ$ 45°



This exploration is scaffolded to allow students to discover the relationships of the sides of special right triangles with their partner rather than being guided through by the teacher. There will be an opportunity to summarize and share your thinking out loud in the next component.



Students may provide different correct answers for question 1a. They may write answers such as “a right triangle.” Display all correct responses on the board until the most precise description of “isosceles right triangle” is determined.



If students get an error message when inputting the expression for $m\angle GTM$ in the calculator, they most likely did not group the denominator when dividing. Ask clarifying questions that lead to the understanding of using the grouping symbol when inputting.



If students answer $HC = \frac{1}{8}$ and/or $WH = \frac{1}{8}$, then they either set up the equation incorrectly, or solved the correct equation incorrectly.

7.6.2 Exploration: Exploring Special Right Triangles (continued)

3. What do you notice about the angles of $\triangle WCH$ and $\triangle MGT$?

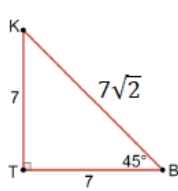
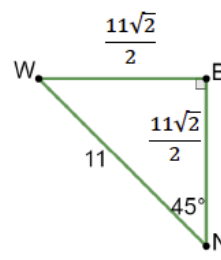
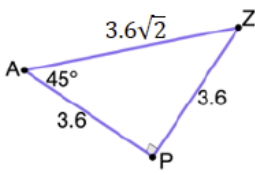
Answers will vary. They both have 45 – 45 – 90 degree angles.

4. What do you notice about the legs of $\triangle WCH$ and $\triangle MGT$?

*Answers will vary.
They both have two legs that are congruent and a hypotenuse that is the product of the leg length times $\sqrt{2}$.*

$\triangle WCH$ and $\triangle MGT$ are 45 – 45 – 90 triangles.

5. The table shows three 45 – 45 – 90 triangles.

$\triangle KBT$	$\triangle WBN$	$\triangle AZP$
		

- a. Discuss with your partner the relationship between the legs of a 45 – 45 – 90 triangle and its hypotenuse. Write a conjecture.

The hypotenuse is equal to the product of the length of a leg and $\sqrt{2}$.



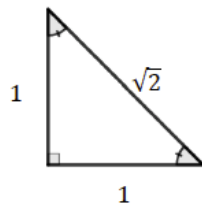
Consider converting questions 3 and 4 to a partner discussion. Students are more likely to share at-length verbally than write in their books. Have them notate their summary in their books for future reference.

7.6.2 Exploration: Exploring Special Right Triangles (continued)

- b. Ask two other pairs of partners for the conjecture they wrote. Listen to their conjecture and write a summary. *You are the only person who should write in the first column of the table.* Have the person initial next to your summary stating that your summary was accurate.

Conjecture	Initials
<i>Answers will vary.</i>	

6. An isosceles right triangle is shown.



Describe how the triangle shows the relationships in a 45 – 45 – 90 triangle.

The legs have the same measurement and the hypotenuse is the leg measurement times $\sqrt{2}$.



Have students try to find other groups with different opinions and approaches to question 5b.



Have students share out their descriptions on question 6 in a whole-class discussion. This is where students may have misconceptions that can be addressed during the discussion.

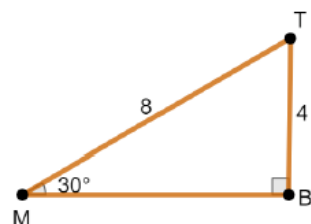
7.6.2 Exploration: Exploring Special Right Triangles (continued)

7. $\triangle MBT$ is a right triangle, where $m\angle MBT = 90^\circ$, $m\angle TMB = 30^\circ$, $TB = 4$, and $MT = 8$.

Find the length of $\overline{MB}$ using the Pythagorean Theorem. Write the answer in simplified radical form.

$$4^2 + MB^2 = 8^2$$

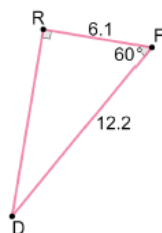
$$MB = 4\sqrt{3}$$



8. $\triangle RFD$ is a right triangle, where $m\angle DRF = 90^\circ$, $m\angle RFD = 60^\circ$, $RF = 6.1$, and $FD = 12.2$.

Discuss with your partner what features $\triangle RFD$ has in common with $\triangle MBT$.

All angle measures are the same and the leg across from the 30° angle is half the length of the hypotenuse.



9. $\triangle MBT$ and $\triangle RFD$ are 30 – 60 – 90 triangles.

The table gives the exact measurements for each of the sides of $\triangle MBT$ and $\triangle RFD$.

Triangle	Short Leg	Long Leg	Hypotenuse
$\triangle MBT$	4	$4\sqrt{3}$	8
$\triangle RFD$	6.1	$6.1\sqrt{3}$	12.2

Complete the table by describing the relationship between the side lengths of a 30 – 60 – 90 triangle.

Sides	Relationship
short leg and hypotenuse	<i>The hypotenuse is double the short leg.</i>
short leg and long leg	<i>The long leg is the product of the short leg times $\sqrt{3}$.</i>



Consider alternative ways of assessing students' understanding by having students explain their thought process for question 7:

- How did you determine the length of $\overline{MB}$?



If students struggle to determine the relationship, provide a different entry point by having the students label the measurement of all missing sides and angles prior to stating what the two triangles have in common.



- Why is knowing that the two triangles are 30-60-90 triangles beneficial?
- How are the sides of one 30-60-90 triangle related to the sides of another 30-60-90 triangle?

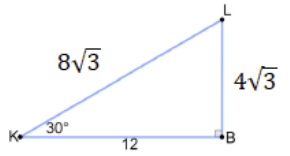
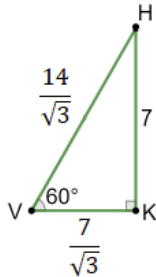
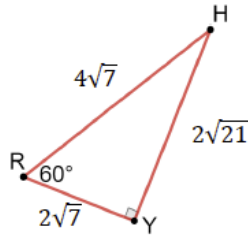
Listen for the correct usage of the vocabulary “similar triangles” and “proportional sides.”



For the table, listen to student descriptions of these relationships for appropriate academic language and specific relationships. Students may list relationships that are accurate but do not connect to the learning outcomes of this lesson.

7.6.2 Exploration: Exploring Special Right Triangles (continued)

10. The table below includes three $30 - 60 - 90$ triangles.

$\triangle KBL$	
$\triangle VHK$	
$\triangle HRY$	



Students may not spend time evaluating these images as there is no “work” required to complete the table. Consider implementing a Notice and Wonder for these images so that students pause here before moving to the discussion part of the next question.

7.6.2 Exploration: Exploring Special Right Triangles (continued)

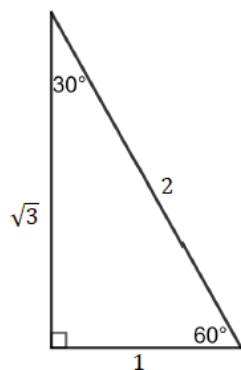
Discuss with your partner the relationships between the sides of a 30 – 60 – 90 triangle in the table. Complete the table by showing how the short side of each of the triangles can be used to determine the long side and the hypotenuse. Write the final length as an exact value.

Triangle	Short Side	Work to Find the Long Side	Work to Find the Hypotenuse
$\triangle KBL$	$4\sqrt{3}$	$\frac{4\sqrt{3} \cdot \sqrt{3}}{4 \cdot 3}$ 12	$\frac{4\sqrt{3} \cdot 2}{8\sqrt{3}}$
$\triangle VHK$	$\frac{7}{\sqrt{3}}$	$\frac{\frac{7}{\sqrt{3}} \cdot \sqrt{3}}{7}$	$\frac{\frac{7}{\sqrt{3}} \cdot 2}{\frac{14}{\sqrt{3}}}$
$\triangle HRY$	$2\sqrt{7}$	$\frac{2\sqrt{7} \cdot \sqrt{3}}{2\sqrt{21}}$	$\frac{2\sqrt{7} \cdot 2}{4\sqrt{7}}$

11. Describe how the triangle shows the relationships in a 30 – 60 – 90 triangle.

The hypotenuse is twice the length of the short side.

The long side is the product of the short side times $\sqrt{3}$.



Consider having students switch partners and discuss their responses to question 10. Students can then return to their original partner and discuss any difference and make corrections.



Listen for student responses to question 11, and their rationale for why this triangle shows the relationships in 30-60-90 triangles. Some students may use surface-level responses instead of using what aligns with the learning outcomes of this activity.

7.6.3 Guided Instruction: 45-45-90 and 30-60-90 Triangles

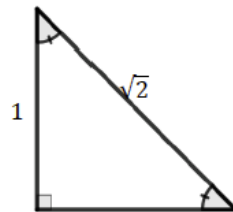
Component Overview: Students will be formally introduced to the side relationships of 45-45-90 and 30-60-90 special triangles.



7.6.3 Guided Instruction: 45-45-90 and 30-60-90 Triangles



The side lengths of a 45 – 45 – 90 triangle can be found using the equation $\text{leg} \cdot \sqrt{2} = \text{hypotenuse}$.



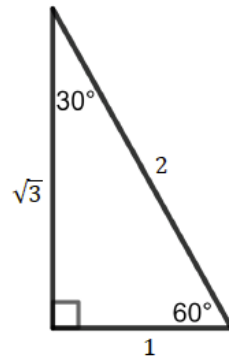
Students may write the exact form of question 1 as $\frac{12}{\sqrt{2}}$ or as $6\sqrt{2}$. Discuss why these are equivalent forms.

1. $\triangle ZYN$ has $m\angle YZN = m\angle YNZ = 45^\circ$, $m\angle ZYN = 90^\circ$, and $ZN = 12$. What are the lengths of the legs of the triangle? Leave your answer as an exact answer.

$$\begin{aligned} \text{leg} \cdot \sqrt{2} &= 12 \\ \text{leg} &= \frac{12}{\sqrt{2}} \end{aligned}$$



The lengths of the short leg and the long leg of a 30 – 60 – 90 triangle can be found using the equation $\text{short leg} \cdot \sqrt{3} = \text{long leg}$.



The lengths of the short leg and the hypotenuse of a 30 – 60 – 90 triangle can be found using the equation $\text{short leg} \cdot 2 = \text{hypotenuse}$.



Why do you think the process for finding the side lengths is different for the two legs of a 30-60-90 special right triangle?

7.6.3 Guided Instruction: 45-45-90 and 30-60-90 Triangles (continued)

2. $\triangle BHK$ is a 30 – 60 – 90 triangle with $m\angle KBH = 30^\circ$, $m\angle BKH = 90^\circ$, and $BK = 5\sqrt{3}$. What are the measures of the other sides of the triangle?

Short Side

$$KH \cdot \sqrt{3} = 5\sqrt{3}$$

$$KH = 5$$

Hypotenuse

$$KH \cdot 2 = BH \quad 10 = BH$$



Encourage students to create a visual representation of this problem to help them see the angles and sides in a way that is consistent with what they have experienced in this and prior activities.

7.6.4 Guided Practice: Solving Special Right Triangles

Component Overview: Students will complete two practice questions with solving special right triangles.



7.6.4 Guided Practice: Solving Special Right Triangles

1. $\triangle LPM$ is a $45 - 45 - 90$ triangle with $m\angle LMP = m\angle MLP = 45^\circ$, $m\angle LPM = 90^\circ$, and $LM = 7$. What are the lengths of $\overline{PM}$ and $\overline{LP}$?

$$\text{leg} \cdot \sqrt{2} = 7$$

$$PM \text{ and } LP = \frac{7}{\sqrt{2}}$$

2. $\triangle TNW$ is a right triangle with $m\angle NTW = 60^\circ$, $m\angle WNT = 90^\circ$, and $TW = 18$. What are the measures of the other sides of the triangle?

$$NT \cdot 2 = 18$$

$$NT = 9$$

$$NT \cdot \sqrt{3} = WN$$

$$9\sqrt{3} = WN$$



Consider pulling together a small group of students who may need additional support based on your observations during the prior activities. Consider working with students on question 1 in small groups and releasing them to try question 2 on their own or in a small group.

7.6.5 Wrap-Up: Ticket Out the Door

Component Overview: Students will complete a Ticket Out the Door on solving a problem involving a special triangle. Students should not use a calculator to complete the wrap-up.



7.6.5 Wrap-Up: Ticket Out the Door

1. $\triangle FHX$ has angle measures $m\angle HXF = 90^\circ$ and $m\angle HFX = 30^\circ$, and side length $HX = 6\sqrt{3}$. What are the exact measures of $\overline{FX}$ and $\overline{FH}$?

$$6\sqrt{3} \cdot \sqrt{3} = FX$$

$$18 = FX$$

$$6\sqrt{3} \cdot 2 = FH$$

$$12\sqrt{3} = FH$$

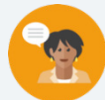


Student responses to this activity can provide data to which components of the lesson, if any, may need additional review.

7.7 The Unit Circle and Its Coordinates

Lesson Introduction

 Benchmarks	MA.912.T.1.1 Define trigonometric ratios for acute angles in right triangles.		MA.912.T.1.2 Solve mathematical and real-world problems involving right triangles using trigonometric ratios and the Pythagorean Theorem.		 Learning Targets	- I can explore relationships between the coordinates of vertices when the triangle sides are in different quadrants. - I can use side ratios of 45-45-90 triangles to solve problems. - I can use side ratios of 30-60-90 triangles to solve problems.
 Benchmark Coherence	Building On			Working Toward		
	MA.8.GR.1.1 Apply the Pythagorean Theorem to solve mathematical and real-world problems involving unknown side lengths in right triangles.			N/A		
 Essential Questions	How do I identify coordinates on the unit circle?					
 Lesson Narrative	This lesson is the first experience students have with the unit circle on a polar graph. Students will work in groups to discover the coordinates on the unit circle by making connections to the side relationships of special right triangles.					
 Component Summary	7.7.1 Warm-Up (~5 min.)	Determining the time for two things to cross Florida (<i>SE p. 408</i>)	7.7.3 Bring It Together (5-10 min.)	Summarizing the findings from the Unit Circle (<i>SE p. 414</i>)		
	7.7.2 Exploration (20-25 min.)	Identifying the coordinates on the Unit Circle (<i>SE p. 408</i>)	7.7.4 Try It (5 min.)	Finding coordinates using the Unit Circle (<i>SE p. 415</i>)		
	7.7.5 Wrap-Up (~5 min.)	Reflecting on understanding and confusions over the Unit Circle (<i>SE p. 415</i>)				
 Resources	Glossary		Materials		Additional Preparation	
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> - Unit Circle - Polar Graph		(Optional) Colored pencils (Optional) Highlighters (Optional) Colored tape		If creating a visual Unit Circle before the lesson, tape the large Unit Circle in a central location in the classroom.	

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may incorrectly identify the x- and y-coordinates or use the incorrect sign. - Students may struggle with working with radicals and recognizing equivalent expressions.	Some calculators may rationalize the denominators in the fractions involving radicals. Consider allowing this lesson to be a non-calculator lesson in order to avoid the confusion, and to monitor student understanding of sine, cosine, and tangent in special right triangles.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Notice and Wonder Students will engage in identifying similarities and differences in question 2a during the 7.7.2 Exploration. Consider adapting this to have students first engage in the Notice and Wonder routine. During 7.7.3 Bring It Together, prompt students with “What do you notice? What do you wonder?” about the triangles shown. Allow students time to think and to write a response before spotlighting different responses.	MA.K12.MTR.5.1: Patterns and Structure The 7.7.2 Exploration is scaffolded for students to write conjectures as they notice patterns and structure in coordinates on a polar graph. Students will bring together their understanding from the exploratory activity in the 7.7.3 Bring It Together, where students will apply the patterns they observed in the 7.7.2 Exploration to find unknown measures and unknown coordinates.
 Differentiate Instruction	Consider taping a large unit circle in a central location where students can identify different angle measures and label the coordinates on the circle. This will allow students to visually see the information while using movement to identify their solution. Throughout the lesson, use question stems to help auditory learners have an entry point when answering questions.	
Lesson Notes:		

7.7.1 Warm-Up: Between Two Numbers

Component Overview: Using the information given, students will determine how long it will take a turtle and a drone to travel across Florida.



7.7.1 Warm-Up: Between Two Numbers



After observing student work, highlight 2-3 students who have solved the problems differently and discuss how each approach is correct.

1. A Florida Box Turtle's average speed is about 0.17 miles per hour.

The North American X-15's top speed is about 163 miles per hour.

How much time would it take for each to travel across Florida, a distance of about 500 miles?

It would take the box turtle about 2941.176 hours or 122.5 days and 3.067 hours.

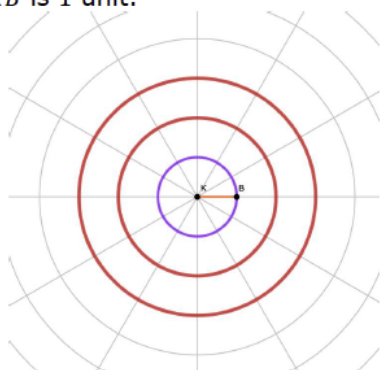
7.7.2 Exploration: The Unit Circle

Component Overview: Students will explore the coordinates of the unit circle using the sine, cosine, and tangent ratios for special right triangles.

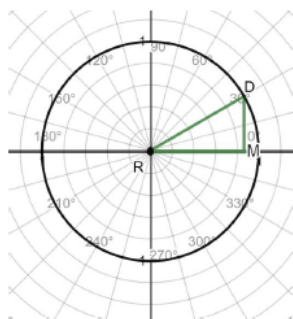
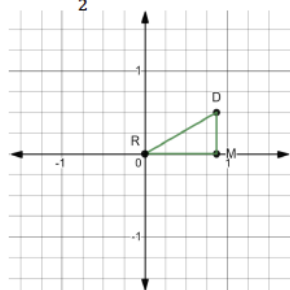


7.7.2 Exploration: The Unit Circle

- Circle K is shown on polar graph paper, where the length of $\overline{KB}$ is 1 unit.



- Draw a circle that has a radius of 2 units.
Answer shown on polar graph.
 - Draw a circle that has a radius of 3 units.
Answer shown on polar graph.
- $\triangle RDM$ is a right triangle formed by drawing a 30° angle at the origin. $\triangle RDM$ is shown on both a coordinate grid and on a polar graph, where $m\angle MRD = 30^\circ$, $m\angle RDM = 60^\circ$, $m\angle DMR = 90^\circ$, $RD = 1$, $DM = \frac{1}{2}$, and $RM = \frac{\sqrt{3}}{2}$.



This Exploration is scaffolded to allow students to work through the problems with their peers. Observe and listen as students work with their partners. Make note of student ideas that you can ask them to share in the whole-group 7.7.3 Bring It Together.



Using no more than 2 minutes, facilitate a discussion regarding the difference between a polar graph and a coordinate graph. Allow students to discuss when a polar graph is useful in real-life scenarios but be mindful of timing.

7.7.2 Exploration: The Unit Circle (continued)

- a. Discuss with your partner any similarities and differences between the graphs. Summarize your discussion.

Answers will vary.

Both graphs have a 30 – 60 – 90 triangle. One is a coordinate grid and the other is a polar graph.

- b. Work with your partner to determine the coordinates of point D .

$$\left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$$

- c. Work with your partner to complete the table.
Leave answers in exact form.

Angle	Cosine	Sine	Tangent
30° ($m\angle R$)	$\frac{\sqrt{3}}{2}$ $\frac{1}{1}$	$\frac{1}{2}$ $\frac{2}{1}$	$\frac{1}{2}$ $\frac{\sqrt{3}}{2}$

- d. Discuss with your partner how the coordinates of point D can be found using the values from the table. Make a conjecture about the relationship between the values of cosine and sine, and the coordinate pair formed by the angle.

Answers will vary. Using the angle located at the origin as the reference angle, the cosine of the reference angle represents the x value of the coordinate point and the sine represents y value of the coordinate point. (cosine, sine)



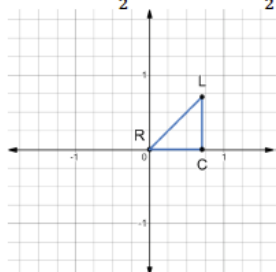
Consider posting a large unit circle with different angle measures listed on the circle. Students can use different tape colors or markings to create the triangles around the center and label the coordinates along the outside of the circle.



How do the side lengths of $\triangle RDM$ relate to special right triangles?

7.7.2 Exploration: The Unit Circle (continued)

3. $\triangle RLC$ is a right triangle formed by drawing a 45° angle at the origin. $\triangle RLC$ is shown on both a coordinate grid and on a polar graph, where $m\angle CRL = 45^\circ$, $m\angle RLC = 45^\circ$, $m\angle RCL = 90^\circ$, $RL = 1$, $RC = \frac{\sqrt{2}}{2}$, and $LC = \frac{\sqrt{2}}{2}$.



- a. Determine the coordinates of point L .

$$\left(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$$

- b. Work with your partner to complete the table.
Leave answers in exact form.

Angle	Sine	Cosine	Tangent
45° ($m\angle R$)	$\frac{\sqrt{2}}{2}$ $\frac{2}{1}$	$\frac{\sqrt{2}}{2}$ $\frac{2}{1}$	$\frac{\sqrt{2}}{2}$ $\frac{2}{2}$

- c. Discuss with your partner whether or not the conjecture that was made about the coordinates of point D could be applied to determine the coordinates of point L . If necessary, revise your conjecture.

Answers may vary. Yes, it can be used. The coordinate point of D and L are both using (cosine, sine).

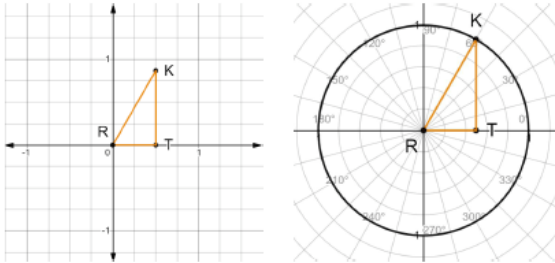


Discussion provides additional entry points for auditory learners. Consider utilizing sentence stems and provide ample time for students to grapple with academic language:

- _____ represents the ____-coordinate, while _____ represents the ____-coordinate.

7.7.2 Exploration: The Unit Circle (continued)

4. $\triangle RKT$ is a right triangle formed by drawing a 60° angle at the origin. $\triangle RKT$ is shown on both a coordinate grid and on a polar graph, where $m\angle TRK = 60^\circ$, $m\angle RKT = 30^\circ$, $m\angle RTK = 90^\circ$, $RK = 1$, $RT = \frac{1}{2}$, and $TK = \frac{\sqrt{3}}{2}$.



- a. Determine the coordinates of point K .

$$\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$$

- b. Work with your partner to complete the table.
Leave answers in exact form.

Angle	Sine	Cosine	Tangent
60° ($m\angle R$)	$\frac{\sqrt{3}}{2}$ $\frac{2}{1}$	$\frac{1}{2}$ $\frac{2}{1}$	$\frac{\sqrt{3}}{2}$ $\frac{1}{2}$

- c. Discuss with your partner how the coordinates will change if the triangle is in a different quadrant. Make a prediction.
Answers will vary. The coordinates will become positive or negative depending on the quadrant.



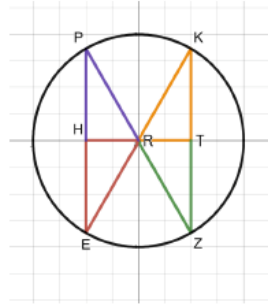
Challenge students to justify why the cosine and sine could be written as $\frac{1}{\sqrt{2}}$ and as $\frac{\sqrt{2}}{2}$, and how this relates to the side relationships of 45-45-90 triangles.



What are the similarities and differences between the triangle in question 4 and the triangle in question 2?

7.7.2 Exploration: The Unit Circle (continued)

5. Four triangles were formed by drawing angles of 60° from the origin. $m\angle TRK = 60^\circ$, $m\angle PRH = 60^\circ$, $m\angle HRE = 60^\circ$, and $m\angle TRZ = 60^\circ$. $RK = 1$, $RP = 1$, $RE = 1$, and $RZ = 1$.



- a. Complete the table.

Point	Coordinates
K	$\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$
P	$\left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$
E	$\left(-\frac{1}{2}, -\frac{\sqrt{3}}{2}\right)$
Z	$\left(\frac{1}{2}, -\frac{\sqrt{3}}{2}\right)$

- b. Points K , P , E , and Z lie on circle R . What is the radius of circle R ?

1

- c. What type of special triangles are $\triangle RKT$, $\triangle RPH$, $\triangle REH$, and $\triangle RZT$?

30-60-90



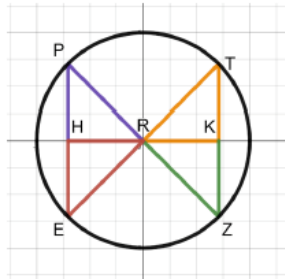
How are sine and cosine related in $\triangle RDM$ and $\triangle KTR$? Why do you think that is?



For kinesthetic and visual learners, give students two different 30-60-90 triangles, and one 45-45-90 triangle. Students can write the side lengths on the triangles with a hypotenuse of 1 and move it around the circle while finding the coordinates.

7.7.2 Exploration: The Unit Circle (continued)

6. Four triangles were formed by drawing angles of 45° from the origin. $m\angle TRK = 45^\circ$, $m\angle PRH = 45^\circ$, $m\angle HRE = 45^\circ$, and $m\angle KRZ = 45^\circ$. $RT = 1$, $RP = 1$, $RE = 1$, and $RZ = 1$.



- a. Complete the table.

Point	Coordinates
T	$\left(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$
P	$\left(-\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$
E	$\left(-\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right)$
Z	$\left(\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right)$

- b. Points T , P , E , and Z lie on circle R . What is the radius of circle R ?

1

- c. What type of special triangles are $\triangle RTK$, $\triangle RPH$, $\triangle REH$, and $\triangle RKZ$?

45 – 45 – 90

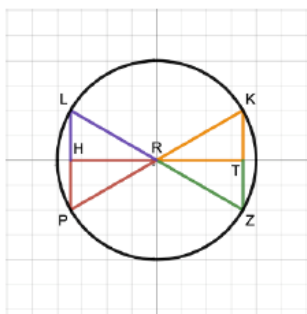


Common Misconception

Students may swap the x - and y -coordinate when listing the coordinates of the points.

7.7.2 Exploration: The Unit Circle (continued)

7. Four triangles were formed by drawing angles of 30° from the origin. $m\angle TRK = 30^\circ$, $m\angle PRH = 30^\circ$, $m\angle HRL = 30^\circ$, and $m\angle TRZ = 30^\circ$. $RK = 1$, $RP = 1$, $RL = 1$, and $RZ = 1$.



Students may find the coordinates to be variations of $(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$, which are equivalent coordinates.

- a. Complete the table.

Point	Coordinates
K	$(\frac{\sqrt{3}}{2}, \frac{1}{2})$
L	$(-\frac{\sqrt{3}}{2}, \frac{1}{2})$
P	$(-\frac{\sqrt{3}}{2}, -\frac{1}{2})$
Z	$(\frac{\sqrt{3}}{2}, -\frac{1}{2})$

- b. Points K , L , P , and Z lie on circle R . What is the radius of circle R ?

1

- c. What type of special triangles are ΔRKT , ΔRPH , ΔRLH , and ΔRZT ?

30 – 60 – 90

7.7.2 Exploration: The Unit Circle (continued)

8. For each type of special triangle, what similarities do the coordinates for each vertex located on the circle share with one another?

Answers will vary. The values of the coordinates all have the same absolute value.

9. For each type of special triangle, what differences do the coordinates for each vertex located on the circle share with one another?

Answers will vary. The coordinates have different signs depending on their location on the coordinate grid.



Notice and Wonder

Before students answer questions 8 and 9, consider utilizing a “Notice and Wonder” routine to offers students the opportunity to make observations regarding the coordinates and to talk through the task to solidify their thinking.

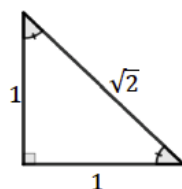
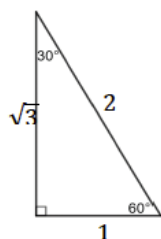
7.7.3 Bring It Together: The Unit Circle

Component Overview: Using the findings from 7.7.2 Exploration, the teacher will discuss how to use the circle to find the coordinates on the circle.



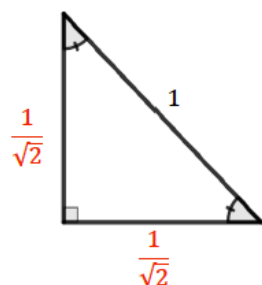
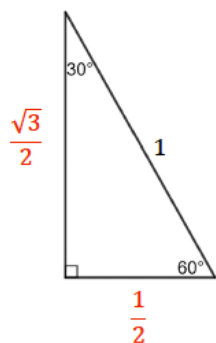
7.7.3 Bring It Together: The Unit Circle

In the Exploration, all of the triangles were one of the two special triangles, $30^\circ - 60^\circ - 90^\circ$ or $45^\circ - 45^\circ - 90^\circ$. The relationships of each of the side lengths for these triangles are shown.



In the circle, the hypotenuse of the triangle was the radius of the circle, which was 1. When a circle has a radius of 1 and is centered at the origin, it is called the unit circle. In the diagrams of the two special triangles the hypotenuse is not 1.

1. Fill in the blanks to show the relationships of the special triangles when the hypotenuse is 1.

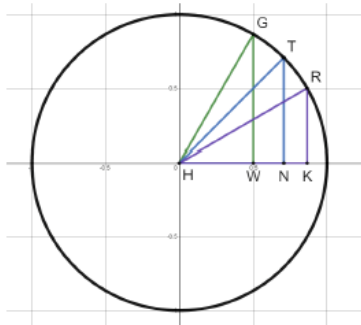


Notice and Wonder

Prompt students with, “What do you notice? What do you wonder?” about the triangles shown. Allow students time to think and to write a response before spotlighting different responses.

7.7.3 Bring It Together: The Unit Circle (continued)

2. A unit circle with $\triangle HKR$, $\triangle HGW$, and $\triangle HNT$ is shown, where $m\angle KHR = 30^\circ$, $m\angle NHT = 45^\circ$, $m\angle WHG = 60^\circ$, $m\angle GWH = 90^\circ$, $m\angle HNT = 90^\circ$, and $m\angle HKR = 90^\circ$.



Allow students to use colored pencils and/or highlighters to color code each reference angle.

Complete the table.

Point	Coordinates
G	$\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$
R	$\left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$
T	$\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$

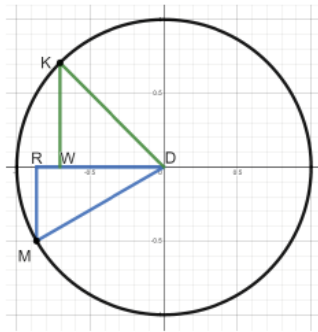
7.7.4 Try It: Finding Coordinates in the Unit Circle

Component Overview: Students work independently to practice using the Unit Circle to find coordinates based on a given reference angle of 30° , 45° , or 60° .



7.7.4 Try It: Finding Coordinates in the Unit Circle

1. A unit circle is shown where $\triangle KDW$ and $\triangle MRD$ are right triangles, $m\angle KDW = 45^\circ$, and $m\angle RDM = 30^\circ$.



Complete the table.

Point	Coordinates
K	$\left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$
M	$\left(-\frac{\sqrt{3}}{2}, -\frac{1}{2}\right)$



Common Misunderstandings

If the signs are incorrect for the coordinates, consider facilitating a discussion over the x - and y -coordinates based on the quadrants. Students can review graphing using On-Ramp to Algebra 1, “Graphing Ordered Pairs.”

If the coordinates of point M are swapped, then the students have a misunderstanding between a 30° and 60° reference angle.

7.7.5 Wrap-Up: Reflection

Component Overview: Students will reflect on their understanding and wonderings from the exploration at the start of the lesson.



7.7.5 Wrap-Up: Reflection

1. Complete the table.

Answers will vary.

The terms and information used in the Exploration that I knew were ...	Something I wondered during the Exploration was ...	Something I learned during the Exploration was ...

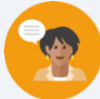


Consider what students write in each column to determine what remediation steps may be needed.

7.8 Using Trigonometric Ratios and the Pythagorean Theorem

Lesson Introduction

 Benchmark	MA.912.T.1.2 Solve mathematical and real-world problems involving right triangles using trigonometric ratios and the Pythagorean Theorem.		 Learning Targets	- I can use trigonometric ratios to solve mathematical problems. - I can use the Pythagorean Theorem to solve mathematical problems.
 Benchmark Coherence	Building On: MA.8.GR.1.1 Apply the Pythagorean Theorem to solve mathematical and real-world problems involving unknown side lengths in right triangles.		Working Toward: N/A	
 Essential Questions	- How do I use trigonometric ratios to solve mathematical problems? - How can I use Pythagorean Theorem to solve mathematical problems?			
 Lesson Narrative	Students are focusing on solving mathematical problems involving right triangles. Students will solve problems that involve using a combination of trigonometric ratios and the Pythagorean Theorem.			
 Component Summary	7.8.1 Warm-Up (~5 min.)	Find side lengths of a right triangle (SE p. 416)	7.8.4 Guided Practice (5 min.)	Solve mathematical problems involving right triangle trigonometry and Pythagorean Theorem (SE p. 419)
	7.8.2 Collaboration (5-10 min.)	Find missing angles and sides in overlapping right triangles (SE p. 417)	7.8.5 Your Turn! (10 min.)	Solve mathematical problems involving right triangle trigonometry and Pythagorean Theorem (SE p. 420)
	7.8.3 Guided Instruction (10-15 min.)	Solve mathematical problems involving right triangle trigonometry and the Pythagorean Theorem (SE p. 418)	7.8.6 Wrap-Up (~5 min.)	Students summarize their learning (SE p. 421)
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> N/A		- Scientific Calculators (Optional) Colored Pencils (Optional) Highlighters	N/A

 Teacher Tips	Common Misconceptions		Things to Consider	
	- Students may incorrectly round when approximating the side length of right triangles using trigonometric ratios. - Students may assume a point is a midpoint in 7.8.5 Your Turn! when solving for the hypotenuse.		- Consider facilitating discussions that allow the students to discuss common misconceptions and how to avoid or overcome these obstacles. - In problems that require several steps, students should not round until the final step. Rounding mid-step could significantly change the answer. If not given a place value for rounding, then determine what place value you would like your students to round to.	
	Literacy and Language Routines		Mathematical Thinking and Reasoning (MTR) Standard	
	Group Presentations During 7.8.2 Collaboration, have students display their work for the class. Choose students to spotlight struggles and strategies used when solving for the angle and segment measures.		MA.K12.MTR.1.1: Participate in Effortful Learning The 7.8.4 Guided Practice and 7.8.5 Your Turn! components require students to think about and discover the correct steps in order to find the final answer. Consider using the language from the benchmark clarifications of MA.K12.MTR.1.1 to help students make the connection between making the commitment to learn and having a positive mindset when solving a challenging task.	
 Differentiate Instruction	Students may struggle with breaking down the information given in each problem. For auditory learners, consider using question stems during 7.8.2 Collaboration and 7.8.3 Guided Instruction to help students make sense of what the problem is giving them and what the problem is asking of them. For visual learners, consider allowing students to color code given information, such as the side lengths of triangles, while solving mathematical problems.			
Lesson Notes:				

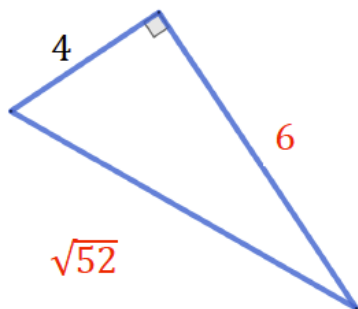
7.8.1 Warm-Up: Critical Thinking

Component Overview: Students are using the digits 0 – 9 to find missing values for the side lengths of a right triangle. Digits used should not be repeated.



7.8.1 Warm-Up: Critical Thinking

1. Use the digits 0-9 one time each to find the missing side lengths of the right triangle. The triangle is not drawn to scale.



Monitor student responses and select a few students with different answers to share how their answer is correct and the strategy used to solve it.

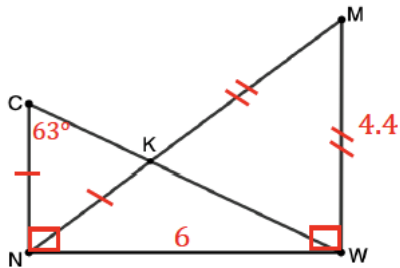
7.8.2 Collaboration: Overlapping Triangles

Component Overview: Students will work in pairs to solve a mathematical problem involving overlapping triangles. Students should use trigonometric ratios and the Pythagorean Theorem to solve for the missing angles and side lengths.

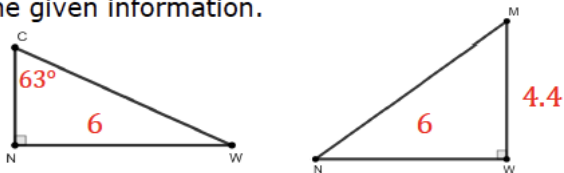


7.8.2 Collaboration: Overlapping Triangles

1. $\triangle MWN$ and $\triangle CNW$ are right triangles that share side $\overline{NW}$ and K is the intersection point of sides $\overline{CW}$ and $\overline{MN}$. In the triangles, the $m\angle MWN = m\angle CNW = 90^\circ$, $MW = 4.4$, $NW = 6$, and $m\angle NCK = 63^\circ$. $\triangle CNK$ is an isosceles triangle with base $\overline{CK}$ and $\triangle MWK$ is an isosceles triangle with base $\overline{WK}$.



- a. Mark the figure with all information given in the problem.
- b. Jasmine redrew the diagram as two separate triangles, as shown. Mark Jasmine's diagram using the given information.



- c. Find the measures of the segments and angles listed in the table. Round to the nearest thousandth. *Answers may vary slightly depending on method and rounding.*

Segment	Length	Angle	Measure
$\overline{MN}$	$MN = 7.44$	$\angle MNW$	36°
$\overline{CN}$	$CN = 3.06$	$\angle WMN$	54°
$\overline{CW}$	$CW = 6.726$	$\angle CWN$	27°
$\overline{KN}$	$KN = 3.06$	$\angle WCN$	63°
$\overline{MK}$	$MK = 4.4$	$\angle CKN$	63°



Allow students to choose their own method of marking given information and congruent parts. Question students about their choice of markings to ensure understanding.

7.8.3 Guided Instruction: Solving Problems With Right Triangles

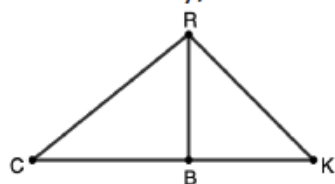
Component Overview: The teacher will use guided instruction to model solving mathematical problems involving trigonometric ratios and the Pythagorean Theorem.



7.8.3 Guided Instruction: Solving Problems With Right Triangles

1. $\triangle RBK$ and $\triangle RBC$ are right triangles that share side $\overline{RB}$. $\angle RBK$ and $\angle RBC$ are both right angles, $m\angle BCR = 38.7^\circ$, $m\angle RKC = 45^\circ$, and $RB = 8$.

- a. Use the given information to complete the table. Whenever possible, give lengths as exact values. If necessary, round to the nearest hundredth.



Segment	Length
$\overline{RC}$	12.80
$\overline{CB}$	9.99
$\overline{RK}$	$8\sqrt{2}$
$\overline{KB}$	8.00
$\overline{CK}$	17.99

- b. What is the area of $\triangle CRK$?

$$A = \frac{1}{2}(17.99)(8) = 71.96 \text{ square units}$$

2. Triangle RGH is shown, where

$m\angle RHG = 90^\circ$,
 $m\angle RWH = 50.2^\circ$,
 $m\angle WRG = 16.5^\circ$,
 $m\angle RGH = 33.7^\circ$, and
 $RH = 12$.

What is the approximate perimeter of $\triangle RWG$? Round your answer to the nearest hundredth.

$$RG: \sin(33.7^\circ) = \frac{12}{RG}; \quad RG \approx 21.63$$

$$WG: \tan(33.7^\circ) = \frac{12}{HG}; \quad HG \approx 17.99$$

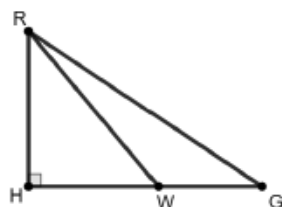
$$\tan(50.2^\circ) = \frac{12}{HW}; \quad HW \approx 9.99$$

$$WG = HG - HW \approx 17.99 - 9.99 \approx 8$$

$$RW: 12^2 + (9.99^2) = RW^2; \quad RW \approx 15.61$$

Therefore, the perimeter of $\triangle RWG$ is:

$$RG + WG + RW \approx 21.63 + 8 + 15.61 \approx 45.24 \text{ units}$$



Common Misconception

Check student answers to ensure that the answers are being rounded correctly. If students are rounding incorrectly, consider facilitating a discussion over how to round.



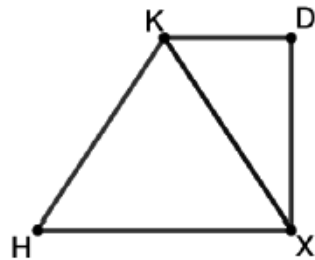
A discussion could provide additional entry points for auditory learners. Consider utilizing sentence stems and provide ample time for students to grapple with academic language:

- I know the altitude of the triangle is _____ because _____.
- I know the base of the triangle is _____.
- Using the area formula for a triangle, I found the area to be _____.

7.8.3 Guided Instruction: Solving Problems With Right Triangles (continued)

3. $\triangle KDX$ and $\triangle HKX$ share side $\overline{KX}$. $\triangle HKX$ is an isosceles triangle with base $\overline{HX}$ and $\triangle KDX$ is a right triangle with $m\angle KDX = 90^\circ$.

If $HX = 13$, $DH = 9$, and $m\angle KXD = 34^\circ$, what is the perimeter of quadrilateral $KDXH$? Round your answer to the nearest hundredth.



First, find KH by finding KX :

$$\cos(34^\circ) = \frac{9}{KX}$$

$$KX = \frac{9}{\cos(34^\circ)} \approx 10.86$$

So, $KH \approx 10.86$.

Next, find KD :

$$\tan(34^\circ) = \frac{KD}{9}$$

$$KD = 9 \cdot \tan(34^\circ)$$

$$KD \approx 6.07$$

$$HK + KD + DX + XH = 10.86 + 6.07 + 9 + 13 = 38.93 \text{ units}$$



For a visual approach to this problem, consider color coding each side using highlighters or colored pencils. This will allow students to track which sides are being solved, as well as which side lengths are congruent. In order to help auditory learners recognize that $\overline{KH}$ and $\overline{KX}$ are congruent, consider using the question stem:

- "I know that _____ and _____ are congruent by the definition of _____."

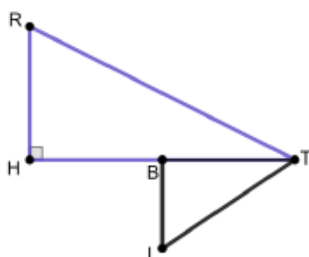
7.8.4 Guided Practice: Right Triangle Trigonometry and the Pythagorean Theorem

Component Overview: The teacher will guide students through the process of solving mathematical problems that involve trigonometric ratios and the Pythagorean Theorem.



7.8.4 Guided Practice: Right Triangle Trigonometry and the Pythagorean Theorem

1. Right triangles RHT and TBL are shown, where B is the midpoint of $\overline{HT}$, $m\angle RHT = m\angle TBL = 90^\circ$, $m\angle RTH = 30^\circ$, $BL = 3$, and $LT = 7$.



- a. What is the length of $\overline{HT}$?

$$4\sqrt{10}$$

- b. What is the area of $\triangle RHT$?

$$\tan(30^\circ) = \frac{RH}{HT}$$

$$\tan(30^\circ) = \frac{RH}{4\sqrt{10}}$$

$$RH = \tan(30^\circ) \cdot 4\sqrt{10} \approx 7.303$$

$$A = \frac{1}{2}(7.303) \cdot 4\sqrt{10}$$

$$A \approx 46.188$$



Allow informal observation throughout the first two activities to determine how “guided” this guided practice should be for you and your classroom. Consider small group instruction for some while others attempt independently, or teacher facilitation of practice with students actively working as a whole group.



This question offers an opportunity to spiral back to special right triangles and the definition of midpoint.

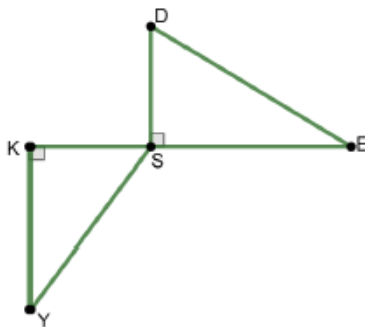
7.8.5 Your Turn!

Component Overview: Students work independently to solve mathematical problems that involve trigonometric ratios and the Pythagorean Theorem.



7.8.5 Your Turn!

1. In the diagram below, triangle DSE and YKS are right triangles that share the vertex S . The $m\angle DSE = m\angle YKS = 90^\circ$, $m\angle DES = 31^\circ$, $m\angle KSY = 53^\circ$, $KE = 37$, and $DS = 10$.



Determine the length of $\overline{KY}$.

$$\tan(31^\circ) = \frac{10}{SE}$$

$$SE = \frac{10}{\tan(31^\circ)}$$

$$SE \approx 16.643$$

$$KS + SE = KE$$

$$KS + 16.643 = 37$$

$$KS \approx 20.357$$

$$\tan(53^\circ) = \frac{KY}{20.357}$$

$$KY = 20.357 \cdot \tan(53^\circ)$$

$$KY \approx 27.015$$



For question 1, students should be instructed to round to the nearest hundredth or write their answer as an exact length. If students are instructed to represent their answer as an exact length, they will not perform any calculations. Students could be asked to show their steps without performing the calculations and then in their final step asked to approximate the length. By doing this, students will demonstrate their understanding in setting up the trig ratios and then show their understanding of how to correctly use their calculator to approximate the solution.

$$\text{Exact length: } \left(\tan 53^\circ \right) \left(37 - \frac{10}{\tan 31^\circ} \right)$$

$$\text{Approximate length: } 27.02$$

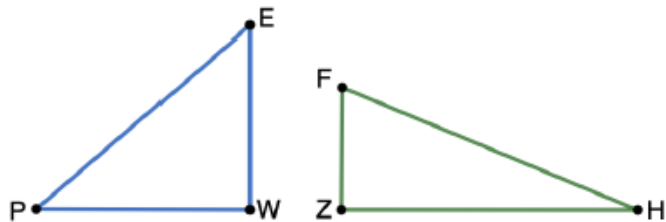


Common Misconception

If students find $KY = 30.74$, then the students assumed that point S was the midpoint of $\overline{KE}$. Consider facilitating a discussion about how information cannot be assumed, unless directly given.

7.8.5: Your Turn! (continued)

2. Triangles PWE and FZH are shown. In $\triangle PWE$, $m\angle WEP = 49.3^\circ$, $PE = 30.6$, and $\angle PWE$ is a right angle.



If $\angle FZH$ is a right angle, FZ is two-thirds the length of EW , and ZH is 1.2 times longer than PW , what is the length of FH ?

$$FZ = \frac{2}{3}EW$$

$$\cos(49.3^\circ) = \frac{EW}{30.6}$$

$$EW = 30.6 \cdot \cos(49.3^\circ)$$

$$EW \approx 19.954$$

$$FZ \approx 13.303$$

$$ZH = 1.2PW$$

$$\sin(49.3^\circ) = \frac{PW}{30.6}$$

$$PW = 30.6 \cdot \sin(49.3^\circ)$$

$$ZH = 1.2(30.6 \cdot \sin(49.3^\circ))$$

$$ZH \approx 27.839$$

$$FH = \sqrt{27.839^2 + 13.303^2}$$

$$FH \approx 30.854$$



For question 2, students should be instructed to round to the nearest hundredth or write their answer as an exact length.



Common Misconception

If students find $PW = 19.95$, then the students used the wrong trigonometric ratio, $\cos 49.3 = \frac{PW}{30.6}$. Have students refer to their notes on the trigonometric ratios and labeling their triangles.

7.8.6 Wrap-Up: Cool Down

Component Overview: Students write a short summary of their learning in the lesson. Student responses can be used to identify misconceptions and plan for upcoming instruction.



7.8.6 Wrap-Up: Cool Down

1. Write a summary of what you learned in this lesson.

Answers will vary.

7.9 Using Trigonometry in the Real World

Lesson Introduction

 Benchmark	MA.912.T.1.2 Solve mathematical and real-world problems involving right triangles using trigonometric ratios and the Pythagorean Theorem		 Learning Target	I can solve real-world problems using trigonometric ratios and the Pythagorean Theorem.
 Benchmark Coherence	Building On: MA.8.GR.1.1 Apply the Pythagorean Theorem to solve mathematical and real-world problems involving unknown side lengths in right triangles.			Working Toward: N/A
 Essential Questions	How can I solve real-world problems using trigonometric ratios and the Pythagorean Theorem?			
 Lesson Narrative	In this lesson, students are introduced to trigonometry in the real world in scenarios involving the angle of depression and the angle of elevation. Students begin by exploring the general concept of the angle of depression and the angle of elevation, prior to learning how to determine the value of those angles to solve real-world problems.			
 Component Summary	7.9.1 Warm-Up (~5 min.)	Notice and Wonder about a spiral staircase (<i>SE p. 422</i>)	7.9.4 Guided Practice (10-15 min.)	Solving using trigonometric ratios with angle of elevation and angle of depression (<i>SE p. 425</i>)
	7.9.2 Collaboration (5 min.)	Exploring angle of depression and angle of elevation (<i>SE p. 422</i>)	7.9.5 Your Turn! (5-10 min.)	Independent practice solving real-world problems using trig (<i>SE p. 426</i>)
	7.9.3 Guided Instruction (5-10 min.)	Solving with angle of elevation and angle of depression (<i>SE p. 423</i>)	7.9.6 Wrap-Up (~5 min.)	Reflecting on gaps from the lesson and strategies to fill them (<i>SE p. 427</i>)
 Resources	Glossary		Materials	
	<u>Key Terms:</u> - Angle of Elevation - Angle of Depression <u>Additional Terms:</u> N/A		- Ruler (optional) String or Yarn	
			Additional Preparation	
			This lesson consists almost entirely of real-world word problems. Consider determining which word problem strategy would best be modeled and implemented with your students.	

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may switch the values of the angle of elevation and the angle of depression. - Students may use the value of the wrong angle as the value of the angle of elevation or the angle of depression.	The structure of the lesson is to have students collaborate to understand the general concept of the angle of depression and the angle of elevation in 7.9.2 Collaboration. Students will learn to find these values in 7.9.3 Guided Instruction. They will apply that, along with previously learned methods, to solve real-world problems for the remainder of the unit.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Notice and Wonder The warm-up provides students with an image of a spiral staircase. They are asked to notate what they notice from the image and what they wonder from the image, along with a more specific wondering about capturing the number of stairs. This routine provides a low entry point for students to engage in general discussion.	MA.K12.MTR.7.1: Real-World Contexts From 7.9.3 Guided Instruction and beyond, students will focus on applying trigonometry to solve real-world problems. Through this, students will connect mathematical concepts to everyday experiences and use models and methods to understand, represent, and solve problems.
 Differentiate Instruction	Physical manipulatives and movement can help students understand angle of elevation and angle of depression. Consider having a student stand on a platform and hold a piece of yarn. Have another student stand on the ground and hold the other end of the yarn. Use this to show where the angle of elevation is and the angle of depression is, so that students have a more engaging visual. This will also help kinesthetic learners by having them interact with angle of elevation and angle of depression. Additional differentiation opportunities are denoted throughout the lesson guide.	
Lesson Notes:		

7.9.1 Warm-Up: Notice and Wonder

Component Overview: Students will describe what they notice and what they wonder about a provided image.



7.9.1 Warm-Up: Notice and Wonder

1. The picture shows the stairs in the Ponce de Leon Inlet Lighthouse.



This activity provides students with an image of a spiral staircase. They are asked to notice what they notice from the image and what they wonder from the image, along with a more specific wondering about capturing the number of stairs. This routine provides a low entry point for students to engage in general discussion.

- a. What do you notice?

Answers will vary. It looks like a shell.

- b. What do you wonder?

Answers will vary. I wonder how many stories there are.

- c. The lighthouse has 203 steps. Do you think this picture includes all of the steps? Explain your answer.

Answers will vary. Yes, I can see all of the steps if I look straight down the center.

7.9.2 Collaboration: Angle of Depression and Angle of Elevation

Component Overview: Students will collaborate to discover the concept of an angle of depression and an angle of elevation. For this activity, students will need a partner or small group.



7.9.2 Collaboration: Angle of Depression and Angle of Elevation

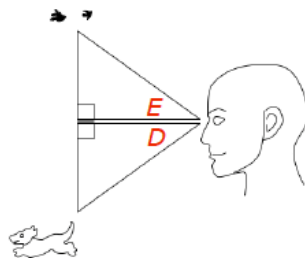
Real-world trigonometric ratio problems involve angles that are made from a horizontal line of sight.

- Using a ruler, draw the horizontal line of sight for the man.



- The horizontal line of sight is always ☐ perpendicular ☒ parallel to the ground.
- The angles created from the horizontal line of sight are called angles of elevation and angles of depression. In the image, Hector is looking at birds in the sky and a dog on the ground.

Label the angle of elevation with an E , label the angle of depression with a D .



For this activity, students will need a ruler in order to complete the step in question 1.



How did you choose which one was the angle of elevation, and the other was the angle of depression?

7.9.2 Collaboration: Angle of Depression and Angle of Elevation (continued)

4. Based on the name of the angle, work with a partner to write a definition for an angle of elevation.

The angle of elevation is the angle created above the horizontal when looking up at an object.

5. Work with a partner to write a definition for an angle of depression.

The angle of depression is the angle created below the horizontal when looking down at an object.

6. Compare your definitions with that of another pair. Discuss any differences. Make changes as necessary.

Answers will vary.



At the conclusion of this activity, consider having students share out their definitions of the angle of elevation and the angle of depression. Record student responses on chart paper or a white board to use as a reference point when they are introduced to the formal definitions and concept in the next activity.

7.9.3 Guided Instruction: Angles of Elevation and Depression

Component Overview: Students will be introduced to the formal definitions of the angle of elevation and the angle of depression and how to use those values to solve a real-world problem.



7.9.3 Guided Instruction: Angles of Elevation and Depression

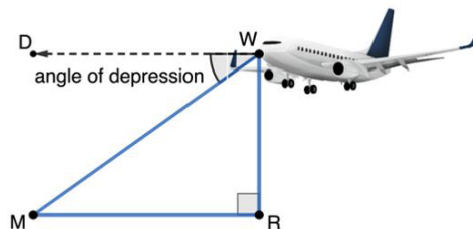


The **angle of elevation** is the angle created above the horizontal when looking up at an object.



The **angle of depression** is the angle created below the horizontal when looking down at an object.

- The angle of depression, $m\angle DWM$, from the airplane to the runway, point M , is 37° .



- Explain why $\angle DWM$ is congruent to $\angle RMW$.
Because $\overline{MR}$ is parallel to $\overline{DW}$ and Alternate Interior Angles are congruent.
- If the airplane is flying at an altitude of 2500 feet, WR , and is descending to the airport, M , what is the distance, MW , the plane has to travel?

$$\sin(37^\circ) = \frac{2500}{MW}; \quad MW \approx 4154.1 \text{ feet}$$

- How far is the runway from a point on the ground directly below the airplane, MR ?
 $(2500)^2 + (MR)^2 = (4154.1)^2$
 $MR = 3317.612 \text{ feet}$



Physical manipulatives and movement can help students understand angle of elevation. Consider having a student stand on a platform and hold a piece of yarn, while another student stands on the ground and holds the other end of the yarn. Use this to show where the angle of elevation and the angle of depression are so that students have a more engaging visual.



Consider modeling for students the labeling of the triangle with the information about the altitude value in question 1b, but releasing them to try and figure out the answer with a partner. Remind them that they have all the information they need to find the correct value. Bring the students back together to review before releasing students to try question 1c on their own.

7.9.3 Guided Instruction: Angles of Elevation and Depression (continued)

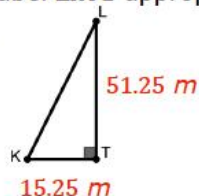
2. Kamala is visiting the Ponce de Leon Inlet Lighthouse. Kamala is standing at point K and is looking at the top of the lighthouse, point L . $\overline{KT}$ is from Kamala's eye level, which is at 1.75 meters (m).



- a. When Kamala looks at the top of the lighthouse, is the angle created an angle of elevation or depression?

Angle of elevation.

- b. The height of the Ponce de Leon Inlet Lighthouse is 53 m. Kamala is standing 15.25 m from the lighthouse. Label $\triangle KTL$ appropriately with this information.



- c. What is the measure of the angle created when Kamala looks at the top of the lighthouse?

$$m\angle K = \tan^{-1}\left(\frac{51.25}{15.25}\right)^\circ \approx 73.429^\circ$$

?

- Why is this an example of an angle of elevation, and not an angle of depression?
- Where would Kamala need to be standing to create an angle of depression?

?

What would be the first step in solving this problem? Why is that first step so important?



Students may need to be reminded to label their triangle with the given information, so they can see that they have the information needed to solve.

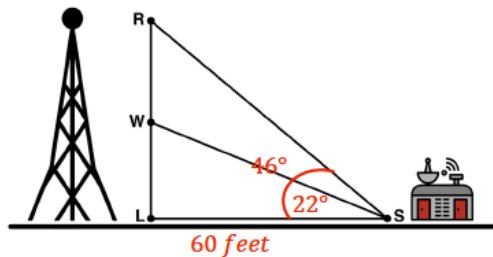
7.9.4 Guided Practice: Trigonometry in the Real World

Component Overview: Students will complete two real-world problems where they will apply what they have learned about trigonometric ratios.



7.9.4 Guided Practice: Trigonometry in the Real World

1. A cell phone tower was built in two sections, with a control shed built 60 feet from the base of the tower on level ground. From the base of the control shed, the angle of elevation, $\angle LSW$, to the top of the first section of tower, $\overline{LW}$, is 22° . The angle of elevation to the top of the second section, $m\angle LSR$, is 46° .



- a. Label the diagram with the information from the problem.
See above.
- b. Determine the length of $\overline{RW}$, the second section of tower. Round your answer to the nearest thousandth of a foot.

$$\tan(22^\circ) = \frac{WL}{60}; \quad WL \approx 24.242 \text{ feet}$$

$$\tan(46^\circ) = \frac{RL}{60}; \quad RL \approx 62.132 \text{ feet}$$

$$RW = RL - WL \approx 37.89 \text{ feet}$$

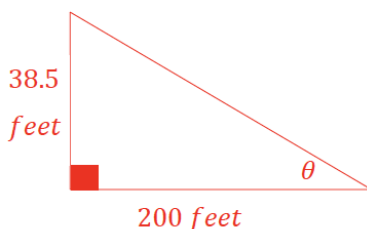


For this activity, consider breaking it up into two problems. Have students complete question 1 independently and then bring them back together. Have students come to the board (or other method) and label one thing in the diagram. Repeat this with multiple students until all labeling has been completed. Then have different students come up to show a step in determining the length. Repeat until all steps have been shown. Engage students in a discussion to better understand any potential misconceptions before moving onto question 2.

7.9.4 Guided Practice: Trigonometry in the Real World (continued)

2. A hot air balloon is flying 38.5 ft above level ground. The Orlando City Soccer field is 200 ft from the point on the ground directly below the balloon.

- a. Draw and label a diagram using the information from the problem.



- b. What is the angle of elevation from the soccer field to the hot air balloon?

$$\theta = \tan^{-1}\left(\frac{38.5}{200}\right)^{\circ} \approx 10.896^{\circ}$$

- c. The hot air balloon ascends further into the air. If the new angle of elevation from the soccer field to the balloon is 17.3° . How many feet did the hot air balloon rise?

$$\tan(17.3)^{\circ} = \frac{\text{height}}{200}; \text{ height} \approx 62.293 \text{ feet}$$

$$62.293 - 38.5 = 23.793$$

The balloon rose an additional 23.793 feet.



For question 2, consider leveraging a Think-Pair-Share instructional routine. Give students a set amount of independent processing time to work through the problem on their own to see what they can do independently. Then call “time,” and have students work together as partners to calibrate their responses and come to a consensus on the correct answers, prior to reviewing as a whole class.

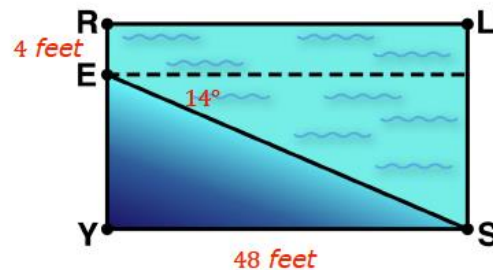
7.9.5 Your Turn!

Component Overview: Students work independently to practice comparing while using trigonometric ratios to solve a real-world problem.



7.9.5 Your Turn!

1. A swimming pool is 48 feet long and 4 feet deep in the shallow end. There is a steady decline of 14° toward the deep end of the pool. Rectangle $RLSY$ and $\triangle SYE$ are shown to model the transition from the shallow end to the deep end. Label the diagram appropriately.



- a. How deep is the pool at the deep end?
 $\tan(14^\circ) = \frac{EY}{48}$, $EY \approx 11.968$ feet
 $LS = RE + EY \approx 15.968$ feet
- b. $\overline{ES}$ represents the slant in the pool from the shallow end to the deep end. What is its length?

$$\cos(14^\circ) = \frac{48}{ES}; \quad ES \approx 49.469 \text{ feet}$$



Consider using this time to pull together a small group of students who may have been struggling during the guided instruction or guided practice portions of this lesson to provide additional support with this problem in a small group setting.



For students who finish early, challenge them to create a word problem that could be solved by using trigonometric ratios. Have them trade their problem with a partner to see if their partner can solve it.

7.9.6 Wrap-Up: Reflection

Component Overview: Students complete a self-assessment based on their level of understanding of using an angle of depression or an angle of elevation.



7.9.6 Wrap-Up: Reflection

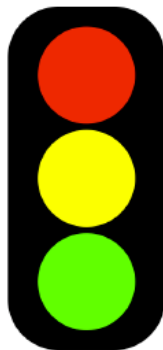
1. Circle the color on the stoplight that best represents your understanding of solving real-world problems using an angle of depression or an angle of elevation.

Answers will vary.

Red: I do not understand.

Yellow: I need some more help.

Green: I got this; I understand!



2. Complete the following statements if you selected red or yellow.

a. I am still unsure about ...

b. To fill in this gap I intend to ...



Consider using student responses from prior self-reflections to consider a student's learning progress. Recording this information and pairing it with assessment data provides insightful feedback for student learning progression.

7.H1 Law of Sines

Lesson Introduction

 Benchmark	MA.912.T.1.3 Apply the Law of Sines and the Law of Cosines to solve mathematical and real-world problems involving triangles.		 Learning Target	I can solve mathematical and real-world problems involving triangles using the Law of Sines.
 Benchmark Coherence	Building On N/A		Working Toward N/A	
 Essential Questions	How can I use the Law of Sines to solve problems?			
 Lesson Narrative	In this lesson, students explore the Law of Sines and how it can be used to solve both mathematical and real-world problems. Students will begin by collaborating to derive the formula for the Law of Sines. They will then learn how to use the Law of Sines as a means to solve problems.			
 Component Summary	7.H1.1 Warm-Up (~5 min.)	Notice and Wonder about two theodolites (<i>SE p. 428</i>)	7.H1.4 Guided Practice (5-10 min.)	Determining measurements using the Law of Sines (<i>SE p. 431</i>)
	7.H1.2 Collaboration (10-15 min.)	Deriving the formula for the Law of Sines (<i>SE p. 428</i>)	7.H1.5 Your Turn! (5 min.)	Determining measurements using the Law of Sines (<i>SE p. 432</i>)
	7.H1.3 Guided Instruction (10 min.)	Determining measurements using the Law of Sines (<i>SE p. 430</i>)	7.H1.6 Wrap-Up (~5 min.)	Reflecting on any difficulties or confusions in the lesson (<i>SE p. 433</i>)
 Resources	Glossary		Materials	
	<u>Key Terms:</u> Law of Sines <u>Additional Terms:</u> N/A		Calculators	
	<u>Additional Preparation</u> This is an Honors lesson. Only students enrolled in Geometry Honors should be required to complete this lesson.			

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may use their calculator incorrectly when finding the sine of an angle as part of the calculation. - Students may cross-multiply the original angle measure and the side length, and then find the sine of that value, instead of finding the sine of the angle first, and then cross-multiplying.	Students often make errors with their calculations by incorrectly using their calculator. Consider the addition of a brief tutorial or think-aloud, showing students how to find the value correctly in their calculator.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
 Differentiate Instruction	Compare and Connect 7.H1.3 Guided Instruction provides students with explicit instruction on the Law of Sines, bridging what students discovered during the prior activity with the tasks they will complete in the next activity. Use this time to demonstrate thinking out loud and prompting students to reflect and respond to that thinking.	MA.K12.MTR.1.1: Participate in Effortful Learning 7.H1.2 Collaboration provides students with an opportunity to derive the Law of Sines. This can potentially cause frustration as it requires students to think deeply and make connections, instead of having the answer up front. Students will be encouraged to build perseverance by modifying methods, as needed, while solving a challenging task.
	Consider creating an anchor chart that displays the Law of Sines, along with a specific example or two of it being applied, as a reference point for students to use as they complete this lesson and the remainder of the unit.	
Additional differentiation opportunities are denoted throughout the lesson guide.		
Lesson Notes:		

7.H1.1 Warm-Up: Notice and Wonder

Component Overview: Students will observe two pictures and identify what they notice and what they wonder.



7.H1.1 Warm-Up: Notice and Wonder

1. A theodolite is an instrument used for measuring angles between designated visible points on the horizontal and vertical planes. Theodolites are often used for land surveying.

Late 16 th century Theodolite	Modern Day Theodolite
	

- a. What do you notice?
Answers will vary. I notice they both sit on a base of some type.
- b. What do you wonder?
Answers will vary. I wonder if the 16th century Theodolite was as accurate as the modern day one?



This activity provides students with an image of a late 16th century theodolite, and a modern-day theodolite. They are asked to notate what they notice from the image and what they wonder from the image. This routine provides a low entry point for students to engage in general discussion.

7.H1.2 Collaboration: Deriving the Formula for Law of Sines

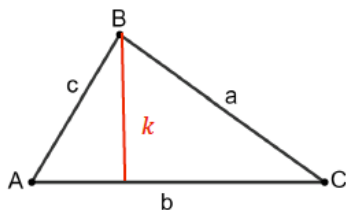
Component Overview: Students will work together to derive the formula for the Law of Sines. For this activity, consider pairing students with a partner or small group.



7.H1.2 Collaboration: Deriving the Formula for Law of Sines

Right triangle trigonometry can be used to solve problems involving right triangles, but not all triangles are right triangles. If a triangle is not a right triangle, the triangle's altitude can be used to create a right triangle so that right triangle trigonometry can be used.

1. $\triangle ABC$ is shown, where $AB = c$, $AC = b$, and $BC = a$.



- a. Use $\triangle ABC$ to do the following.

- Sketch an altitude from vertex B .
- Label the altitude k .

The altitude creates two right triangles inside $\triangle ABC$. Notice that $\angle A$ is contained in one of the right triangles, and $\angle C$ is contained in the other.

- b. Complete each equation using right triangle trigonometry.

$$\sin(A) = \frac{k}{c}$$

$$\sin(C) = \frac{k}{a}$$

- c. Discuss with your partner why both equations use k . Summarize your discussion.

Answers will vary. Both equations use side k because it is shared between both triangles.



This activity provides students with an opportunity to collaborate on deriving the Law of Sines. This can potentially cause frustration as it requires students to think deeply and make connections, instead of having the answer up front. Students will be encouraged to build perseverance by modifying methods, as needed, while solving a challenging task.

7.H1.2 Collaboration: Deriving the Formula for Law of Sines (continued)

- d. Solve each equation for k .

$$k = c \sin(A), \quad k = a \sin(C)$$

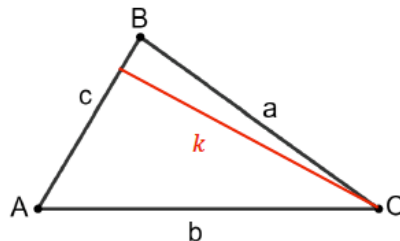
- e. Set the equations equal to each other to form a new equation.

$$c \sin(A) = a \sin(C)$$

- f. Rewrite the new equation by regrouping a with $\sin(A)$ and c with $\sin(C)$.

$$\frac{\sin(A)}{a} = \frac{\sin(C)}{c}$$

2. $\triangle ABC$ is shown where $AB = c$, $AC = b$, and $BC = a$.



- a. Use $\triangle ABC$ to do the following.

- Sketch an altitude from vertex C .
- Label the altitude k .

The altitude creates two right triangles inside $\triangle ABC$. Notice that $\angle A$ is contained in one of the right triangles, and $\angle B$ is contained in the other.



A common error in question 1d is for students to place the value of c (or a) inside the parenthesis with the sine command, creating something that looks like $k = \sin(A \cdot c)$, or $k = \sin(C \cdot a)$, which will produce the incorrect value. If you see this error, remind students of what it looks like in a non-sine environment, using a coefficient without the sine command.



Question 1f can be a confusing instructional request for students. Help them understand what this is asking, if they appear lost when trying to determine their answer.

7.H1.2 Collaboration: Deriving the Formula for Law of Sines (continued)

- b. Complete each equation using right triangle trigonometry.

$$\sin(A) = \frac{k}{b}$$

$$\sin(B) = \frac{k}{a}$$

- c. Solve each equation for k .

$$k = b \sin(A), \quad k = a \sin(B)$$

- d. Set the equations equal to each other to form a new equation.

$$b \sin(A) = a \sin(B)$$

- e. Rewrite the new equation by regrouping a with $\sin(A)$ and b with $\sin(B)$.

$$\frac{\sin(A)}{a} = \frac{\sin(B)}{b}$$



The **Law of Sines** is $\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$

The Law of Sines may also be written as

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$



- Why is k used in both equations on question 2b?
- Is there a scenario where it would only be used in one?



Why do the two equations need to be set equal to each other?



Consider creating an anchor chart that displays the Law of Sines, along with a specific example or two of it being applied, as a reference point for students to use as they complete this lesson and the remainder of the unit.

7.H1.3 Guided Instruction: Law of Sines

Component Overview: Students will be formally introduced to the Law of Sines and its use in two problems.



7.H1.3 Guided Instruction: Law of Sines

- Some of the measurements of $\triangle DLC$ are $DL = 17$, $m\angle DLC = 40^\circ$, and $m\angle CDL = 75^\circ$. Determine the measurement of each part of $\triangle DLC$. Round to the nearest tenth.

Part	Measurement
$\overline{LC}$	$LC = 18.1$
$\overline{DC}$	$DC = 12.1$
$\angle DCL$	$m\angle DCL = 65^\circ$



This activity provides students with explicit instruction on the Law of Sines. It bridges what students discovered during the activity in the prior activity together with the tasks they will complete in the next activity. Use this time to demonstrate thinking out loud and prompting students to reflect and respond to that thinking.

- In $\triangle RFM$, $RM = 66$, $FM = 30$, and $m\angle RFM = 26^\circ$. Determine the measurement of each part of $\triangle RFM$. Round to the nearest tenth.

Part	Measurement
$\overline{FR}$	$FR = 91.6$
$\angle FRM$	$m\angle FRM = 11.5^\circ$
$\angle FMR$	$m\angle FMR = 142.5^\circ$



Do you find the use of the Law of Sines challenging? Why or why not?

7.H1.4 Guided Practice: Law of Sines

Component Overview: Students will practice two problems on the use of the Law of Sines.



7.H1.4 Guided Practice: Law of Sines

1. $\triangle VY$ has side measurements of $VY = 8.5$ and $LY = 16.5$. The measure of $\angle LVY$ is 84° . Find all missing sides and angles. Round to the nearest tenth.

$$\frac{\sin(84)}{16.5} = \frac{\sin(L)}{8.5}; \quad m\angle VLY = 30.8^\circ$$

$$180 - 30.8 - 84 = 65.2^\circ; \quad m\angle LYV = 65.2^\circ$$

$$\frac{\sin(84)}{16.5} = \frac{\sin(65.2)}{LV}; \quad LV = 15.1 \text{ units}$$

2. The three known measurements of $\triangle DNH$ are $DN = 26$, $m\angle NDH = 80^\circ$, and $m\angle DNH = 75^\circ$. Find all missing sides and angles. Round to the nearest tenth.

$$180 - 80 - 75 = 25^\circ; \quad m\angle DHN = 25^\circ$$

$$\frac{\sin(25)}{26} = \frac{\sin(80)}{NH}; \quad NH = 60.6 \text{ units}$$

$$\frac{\sin(25)}{26} = \frac{\sin(75)}{DH}; \quad DH = 59.4 \text{ units}$$



For this activity, consider breaking it up into two problems. Have students complete question 1 independently, and then bring them back together. Have each student describe a step of the process. Repeat until all steps have been shown. Engage students in a discussion to better understand any potential misconceptions before moving on to question 2.



For question 2, consider leveraging a Think-Pair-Share instructional routine. Give students a set amount of independent processing time to work through the problem on their own to see what they can do independently. Then, call "time," and have students work together as partners to calibrate their responses and come to a consensus on the correct answers, prior to reviewing as a whole class.

7.H1.5 Your Turn!

Component Overview: Students work independently to practice using the Law of Sines to solve a mathematical and a real-world problem.

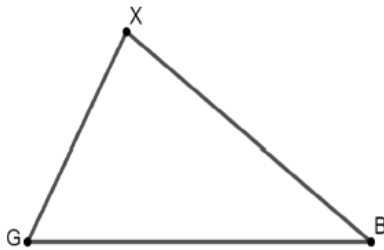


7.H1.5 Your Turn!

1. In $\triangle LSK$, $LS = 7$, $m\angle KLS = 31^\circ$, and $m\angle KSL = 107^\circ$. Determine the length of $\overline{KS}$. Round to the nearest tenth.

$$\frac{\sin(31)}{KS} = \frac{\sin(42)}{7}; \quad KS = 5.4 \text{ units}$$

2. Cell phone towers represented by points G and B are 25 miles apart. A new tower, X , is being built so that $m\angle G = 54^\circ$ and $m\angle B = 26^\circ$. How far from tower G will tower X be built? Round to the nearest tenth.



$$\frac{\sin(100)}{25} = \frac{\sin(26)}{GX}; \quad GX = 11.1 \text{ miles}$$



Consider using this time to pull together a small group of students who may have been struggling during the Guided Instruction or Guided Practice activities of this lesson to provide additional support with this problem in a small group setting.



How do you think a cell phone company can benefit from having the information presented in question 2?

7.H1.6 Wrap-Up: Reflection

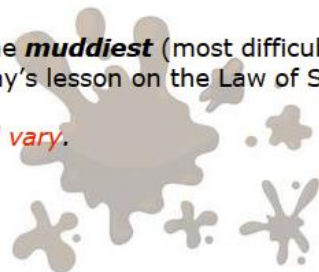
Component Overview: Students complete a self-assessment of their muddiest (most difficult or confusing) point about the Law of Sines.



7.H1.6 Wrap-Up: Reflection

1. What was the **muddiest** (most difficult or confusing) point in today's lesson on the Law of Sines?

Answers will vary.



Consider pairing students to discuss their reflections to see if peers can answer any of their questions.

7.H2 Law of Cosines

Lesson Introduction

 Benchmark	MA.912.T.1.3 Apply the Law of Sines and the Law of Cosines to solve mathematical and real-world problems involving triangles.		 Learning Target	I can solve mathematical and real-world problems that involve triangles using the Law of Cosines.
 Benchmark Coherence	Building On		Working Toward	
	N/A		N/A	
 Essential Questions	How can you solve mathematical and real-world problems that involve triangles using the Law of Cosines?			
 Lesson Narrative	In this lesson, students will learn about the Law of Cosines and how it can be used to solve both mathematical and real-world problems. Students will begin by collaborating to derive the formula for the Law of Cosines. They will then learn how to use the Law of Cosines as a means to solve problems.			
 Component Summary	7.H2.1 Warm-Up (~5 min.)	Reasoning if two triangles are similar (<i>SE p. 434</i>)	7.H2.4 Guided Practice (10 min.)	Finding measures using the Law of Cosines (<i>SE p. 437</i>)
	7.H2.2 Collaboration (10-15 min.)	Deriving the formula for the Law of Cosines Collaboration activity (<i>SE p. 434</i>)	7.H2.5 Your Turn! (5-10 min.)	Finding measures using the Law of Cosines (<i>SE p. 438</i>)
	7.H2.3 Guided Instruction (~10 min.)	Finding measures using the Law of Cosines (<i>SE p. 436</i>)	7.H2.6 Wrap-Up (~5 min.)	Students summarize their learning (<i>SE p. 439</i>)
 Resources	Glossary		Materials	
	<u>Key Terms:</u> Law of Cosines <u>Additional Terms:</u> N/A		Calculators	
			Additional Preparation This is an Honors lesson. Only students enrolled in Geometry Honors should be required to complete this lesson.	

 Teacher Tips	Common Misconceptions - Students may not apply the order of operations correctly when evaluating the $2ab \cos(C)$, and accidentally take the cosine of the wrong value. - Students may use the Law of Sines instead of the Law of Cosines.	Things to Consider Students often make errors with their calculations by incorrectly using their calculator. Consider the addition of a brief tutorial or think-aloud, showing students how to find the value correctly in their calculator.
	Literacy and Language Routines Think-Pair-Share 7.H2.2 Collaboration provides students with an opportunity on deriving the Law of Cosines. Students will work together using this instructional routine. Students will have the opportunity to work independently before sharing with a partner in a collaborative discussion. Students will then prepare to share their discussion with the class.	Mathematical Thinking and Reasoning (MTR) Standard MA.K12.MTR.5.1: Patterns and Structure During this lesson, students will discover the Law of Cosines and how to use it to solve mathematical and real-world problems. Through the lesson activities, students will need to focus on relevant details within a problem, and create plans and procedures to logically order steps to solve problems.
	 Differentiate Instruction Consider your groupings for the 7.H2.2 Collaboration carefully. Use the performance data from the prior lesson to determine groupings, as this Collaboration activity will unfold in a similar structure as the one in the previous lesson.	
Lesson Notes:		

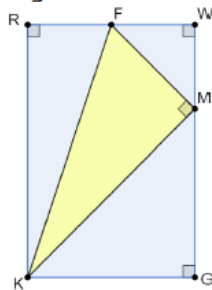
7.H2.1 Warm-Up: Reasoning With Triangles

Component Overview: Students will review a diagram of a triangle inscribed in a rectangle and answer two questions, including evaluating arguments.



7.H2.1 Warm-Up: Reasoning with Triangles

1. A diagram of a triangle inscribed in a rectangle is shown.



This activity provides students with an opportunity to engage in MA.K12.MTR.4.1: Discussion and Reflection. Students will get to analyze the mathematical thinking of others, justify results by explaining methods and processes, and construct possible arguments based on evidence.

- a. Liu Yang told her partner that all of the right triangles in the diagram are similar. Determine if the diagram contains enough information to verify Liu Yang's statement. Explain your reasoning.
Answers will vary. The diagram does not contain enough information to determine if the triangles are similar. In order to determine similarity, the image would have to show an additional angle for each image or proportional side lengths.
- b. In the diagram, $m\angle RFK = 72^\circ$ and $m\angle MKF = 63^\circ$. Use these measures to evaluate your reasoning. If necessary, make adjustments.
With this given information, $\triangle FWM \sim \triangle MGK$ as they both have angle measures of 90° , 81° , and 9° . The other two triangles pictured have different angle measures and are therefore not similar. Liu Yang's statement is still incorrect that all of the right triangles in the diagram are similar.

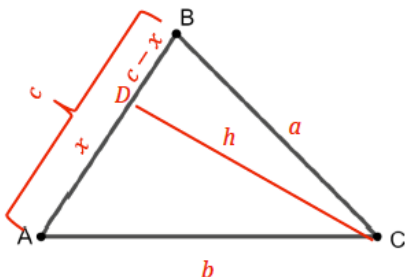
7.H2.2 Collaboration: Deriving the Law of Cosines

Component Overview: Students will work together with a partner or a small group to derive the formula for the Law of Cosines.



7.H2.2 Collaboration: Deriving the Law of Cosines

1. A triangle is shown below.



- a. Use $\triangle ABC$ to complete the following.

- Sketch an altitude from vertex C .
- Label the altitude h , and label the point on $\overline{AB}$ as D .
- Label each line segment. $AC = b$, $AD = x$, $CB = a$, $DB = c - x$, $AB = c$

The altitude creates two right triangles inside $\triangle ABC$. Notice that $\angle A$ is contained in one of the right triangles, and $\angle B$ is contained in the other.

- b. Use the labels on the two right triangles inside $\triangle ACB$ to complete the table.

$\triangle ADC$	$\triangle CDB$
$x^2 + h^2 = b^2$	$h^2 + (c - x)^2 = a^2$

- c. What do you notice about both equations?

Both equations have the variable term h^2 .



This activity provides students with an opportunity to collaborate on deriving the Law of Cosines. This can potentially cause frustration, as it requires students to think deeply and make connections, instead of having the answer up front. Students will be encouraged to build perseverance by modifying methods as needed while solving a challenging task.



What is the purpose of sketching an altitude as part of this problem?

7.H2.2 Collaboration: Deriving the Law of Cosines (continued)

- d. Rearrange the Pythagorean Theorem for $\triangle ADC$ in terms of h^2 .

$$h^2 = b^2 - x^2$$

- e. Substitute the new equation in place of h^2 in $\triangle CDB$.

$$b^2 - x^2 + (c - x)^2 = a^2$$

- f. Use cosine to write the trigonometric ratio for $\angle A$.

$$\cos A = \frac{x}{b}$$

- g. Solve the trigonometric ratio for $\angle A$ for x .

$$x = b(\cos A)$$

- h. Substitute $b \cos A$ in place of x in the equation from part e.

$$b^2 - (b(\cos A))^2 + (c - (b(\cos A)))^2 = a^2$$

- i. Expand and combine like terms.

$$a^2 = b^2 + c^2 - 2bc(\cos A)$$



Students who struggle with questions 1d through 1i may need additional support with literal equations/rearranging formulas from prior units.



Students may not recall what it means to "expand." Consider this misconception when circulating during this activity.

If students struggle with expanding from question 1h to question 1i, consider having them expand and combine like terms in question e before substituting in for question 1h.

7.H2.3 Guided Instruction: Law of Cosines

Component Overview: Students will be formally introduced to the Law of Cosines and its use in two, multi-part problems.



7.H2.3 Guided Instruction: Law of Cosines



The **Law of Cosines** can be written as one of the following:

$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$b^2 = a^2 + c^2 - 2ac \cos B$$

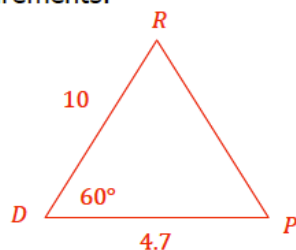
$$c^2 = a^2 + b^2 - 2ab \cos C$$



Consider creating an anchor chart that displays the Law of Cosines, along with a specific example or two of it being applied as a reference point for students to use as they complete this lesson and the remainder of the unit.

1. In $\triangle DRP$, $DR = 10$, $DP = 4.7$, and $m\angle RDP = 60^\circ$.

- a. Draw and label a triangle with the given measurements.



- b. Use the law of cosines to find the measure of $\overline{RP}$.

$$(RP)^2 = 10^2 + 4.7^2 - 2(10)(4.7)(\cos 60)$$

$$RP \approx 8.665$$

- c. Use the law of cosines to find $m\angle RPD$.

$$10^2 = 4.7^2 + 8.665^2 - 2(4.7)(8.665)(\cos P)$$

$$m\angle RPD \approx 91.990^\circ$$



This activity provides students with explicit instruction on the Law of Cosines, bridging what students discovered in the prior component with the tasks they will complete in the next component. Use this time to demonstrate thinking out loud, and prompt students to reflect and respond to that thinking.

7.H2.3 Guided Instruction: Law of Cosines (continued)

2. The measurements of the sides of $\triangle RLP$ are $RL = 4.5$, $PL = 7.2$, and $RP = 6.7$. Determine the measure of the angles of $\triangle RLP$.

Angle	Measurement
$\angle L$	65.181°
$\angle P$	37.561°
$\angle R$	77.258°

$$7.2^2 = 4.5^2 + 6.7^2 - 2(4.5)(6.7) \cdot \cos(R)$$

$$m\angle R \approx 77.258^\circ$$

$$6.7^2 = 4.5^2 + 7.2^2 - 2(4.5)(7.2) \cdot \cos(L)$$

$$m\angle L \approx 65.181^\circ$$

$$180 - (77.258 + 65.181) = 37.561$$

$$m\angle P \approx 37.561^\circ$$



Do you find the use of the Law of Cosines challenging? Why or why not?

7.H2.4 Guided Practice: Law of Cosines

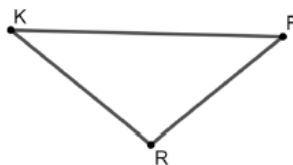
Component Overview: Students will practice two problems on the use of the Law of Cosines.



7.H2.4 Guided Practice: Law of Cosines

1. $\triangle KFR$ is shown. $KF = 15$, $FR = 9$, and $KR = 10$. Find the missing angles.

Angle	Measurement
$\angle F$	40.274°
$\angle K$	35.577°
$\angle R$	104.149°



$$15^2 = 10^2 + 9^2 - 2(10)(9)(\cos(R))$$

$$m\angle R \approx 104.149^\circ$$

$$10^2 = 15^2 + 9^2 - 2(15)(9)(\cos(F))$$

$$m\angle F \approx 40.274^\circ$$

$$180 - (104.149 + 40.274) = 35.577$$

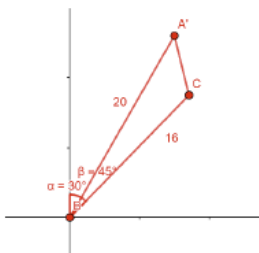
$$m\angle K \approx 35.577^\circ$$

2. Two fishing boats begin traveling in the same direction from a port in Miami. One travels 20 nautical miles per hour on a bearing of 30° and the other travels 16 nautical miles per hour on a bearing of 45° . Assuming each boat maintains its course and speed, how far apart will the fishing boats be after three hours?

$$c^2 = 20^2 + 16^2 - 2(20)(16)(\cos(15^\circ))$$

$$c \approx 6.149$$

$$3c \approx 18.447 \text{ miles apart}$$



Allow informal observation throughout the previous two components to dictate how “guided” this Guided Practice should be for you and your classroom. Consider small group instruction for some while others attempt independently, or consider teacher facilitation of practice with students actively working as a whole group.



Some students may struggle with question 2 because no visual is provided and there is a lot of information in the question. Encourage students to create a triangle and label the information provided to use as a visual for visual learners. Consider using the MLR 6: “Three Reads” routine for auditory learners to make sense of the question.

7.H2.5 Your Turn!

Component Overview: Students work independently to practice using the Law of Cosines to solve two mathematical problems.



7.H2.5 Your Turn!

1. In $\triangle BNW$, $BN = 9$, $NW = 12$, and $BW = 13$. Find the missing angles.

Angle	Measurement
$\angle B$	63.064°
$\angle N$	74.974°
$\angle W$	41.962°

$$13^2 = 9^2 + 12^2 - 2(9)(12)(\cos(N))$$

$$m\angle N \approx 74.974^\circ$$

$$12^2 = 13^2 + 9^2 - 2(13)(9)(\cos(B))$$

$$m\angle B \approx 63.064^\circ$$

$$180 - (75 + 63.1) = 41.962$$

$$m\angle W \approx 41.962^\circ$$

2. In $\triangle SLC$, $SC = 34$, $SL = 22$, and $m\angle LSC = 42^\circ$. Find all missing sides and angles.

Part	Measurement
$\overline{LC}$	22.984
$\angle C$	39.828°
$\angle L$	98.172°

$$(LC)^2 = 22^2 + 34^2 - 2(22)(34)(\cos(42))$$

$$(LC)^2 \approx 528.255$$

$$LC \approx 22.984$$

$$22^2 = 34^2 + 22.984^2 - 2(34)(22.984)(\cos(C))$$

$$m\angle C \approx 39.828^\circ$$

$$180 - (42 + 39.828) = 98.172$$

$$m\angle L \approx 98.172^\circ$$



Encourage students to create a visual by labeling the information given. This will allow students to see the connection between each given side length, and each missing angle.



- What are some similarities that you have noticed between the Law of Sines and the Law of Cosines?
- What are some of the differences?

7.H2.6 Wrap-Up: Cool Down

Component Overview: Students write a short summary of their learning in the lesson. Student responses can be used to identify misconceptions and plan for upcoming instruction.



7.H2.6 Wrap-Up: Cool Down

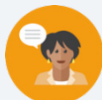
1. Write a summary of what you learned today.

Answers will vary.

7.H3 Area of a Triangle Using the Sine Ratio

Lesson Introduction

 Benchmark	MA.912.T.1.4 Solve mathematical problems involving finding the area of a triangle given two sides and the included angle.		 Learning Target	I can find the area of a triangle using the sine ratio.
 Benchmark Coherence	Building On N/A		Working Toward N/A	
 Essential Questions	How can I find the area of a triangle using the sine ratio?			
 Lesson Narrative	In this lesson, students work to solve mathematical problems of finding the area of a triangle, given two sides and the included angle, using the sine ratio. Students will begin by exploring the area of a triangle using trigonometry. They will then formally learn the three different formulas for finding the area of a triangle using sine.			
 Component Summary	7.H3.1 Warm-Up (~5 min.)	Observations of mathematics everywhere (<i>SE p. 440</i>)	7.H3.3 Guided Practice (10-15 min.)	Finding the area of a triangle using trigonometry (<i>SE p. 442</i>)
	7.H3.2 Exploration (15 min.)	Deriving the formulas for using the sine ratio to find the area of a triangle (<i>SE p. 440</i>)	7.H3.4 Your Turn! (10-15 min.)	Finding the area of a triangle using trigonometry (<i>SE p. 442</i>)
	7.H3.5 Wrap-Up (~5 min.)	Students summarize their learning (<i>SE p. 443</i>)		
 Resources	Glossary		Materials	
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> N/A		Calculators	
		Additional Preparation		
		This is an Honors lesson. Only students enrolled in Geometry Honors should be required to complete this lesson.		

 Teacher Tips	Common Misconceptions - Students may mistakenly invert the values in the sine ratio. - Students may not use the order of operations correctly when evaluating the formulas for finding the area of a triangle using sine.	Things to Consider The purpose of 7.H3.2 Exploration is for students to connect what they know about finding the area of a triangle, given all the necessary side information, to situations where two sides and the included angle are given. Try not to give too much away during the Exploration, so students can make the appropriate connections.
	Literacy and Language Routines Take Turns This lesson does not have a Guided Instruction. Consider leveraging the “Take Turns” instructional routine during the 7.H3.3 Guided Practice. This way, students can take turns in the work of the activity, being able to explain the rationale for their choices, and listen to another’s rationale to deepen their understanding.	Mathematical Thinking and Reasoning (MTR) Standard MA.K12.MTR.5.1: Patterns and Structure During this lesson, students will learn to connect what they have learned about finding the area of a triangle to finding the area of a triangle using trigonometry. Students will relate those previously learned concepts to new concepts while looking for similarities among problems.
	 Differentiate Instruction Consider adding in a flashcard or rapid-fire activity for students. Create pictures of triangles where the area could be solved using a basic area formula and triangles where students would need to use trigonometry. This can help students quickly recognize when one method would be required versus another. Additional differentiation opportunities are denoted throughout the lesson guide.	
Lesson Notes:		

7.H3.1 Warm-Up: History of Mathematics

Component Overview: Students will review information about Hannah Fry, an English mathematician, and respond to two related questions.



7.H3.1 Warm-Up: History of Mathematics

Hannah Fry is an English mathematician who works alongside physicists, mathematicians, computer scientists, architects, and geographers to study the patterns in human behavior. Fry's research applies to a wide range of social problems and questions, from shopping and transport to urban crime, riots, and terrorism.

Fry has also written two popular math books: *The Mathematics of Love: Patterns, Proofs, and the Search for the Ultimate Equation* and *The Indisputable Existence of Santa Claus*.

Fry is quoted as saying, "There is a mathematical angle to almost anything."

1. Interpret this quote in your own words.

Answers will vary. You can find a way make almost anything into a story problem.

2. What is a specific real-world example where mathematics is present?

Answers will vary. Finding the area of a room to carpet it.



Consider having students do additional reading and/or research on Hannah Fry as a way to enhance this activity.

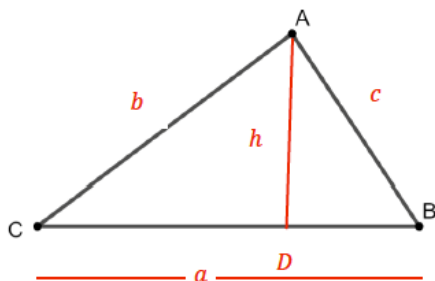
7.H3.2 Exploration: Area of a Triangle

Component Overview: Students will explore the area of a triangle. Students will connect situations where they can solve using a basic area formula to where the use of trigonometry would be required.



7.H3.2 Exploration: Area of a Triangle

1. $\triangle ABC$ is shown.



a. Use $\triangle ABC$ to do the following.

- Draw an altitude from point A to side $\overline{CB}$
- Label the altitude as h .
- Label the point where the altitude intersects side $\overline{BC}$ as D .
- Label the sides as $AB = c$, $BC = a$, and $AC = b$.

The altitude creates two right triangles inside $\triangle ABC$.

b. Write the equation that can be used to find the area of $\triangle ABC$.

$$A = \frac{1}{2}ah$$

c. Complete each equation using right triangle trigonometry.

$\triangle ADB$	$\triangle ADC$
$\sin(B) = \frac{h}{c}$	$\sin(C) = \frac{h}{b}$



The purpose of the Exploration activity is for students to connect what they know about finding the area of triangle, given all the necessary side information, to situations where two sides and the included angle are given. Try not to give too much away during the Exploration, so students can make the appropriate connections.



- How did you determine the ratio for question 1c?
- How can you verify that you are correct?

7.H3.2 Exploration: Area of a Triangle (continued)

- d. Rewrite each equation in terms of h .

$\triangle ADB$	$\triangle ADC$
$h = c \sin B$	$h = b \sin C$

- e. Substitute each expression into the formula,
 $A = \frac{1}{2}ah$.

$\triangle ADB$	$\triangle ADC$
$A = \frac{1}{2}ac \sin B$	$A = \frac{1}{2}ab \sin C$

- f. What do you notice about the two equations?
Answers will vary. Both equations have $\frac{1}{2}a$.

2. $\triangle ABC$ is shown.

- a. Find the area of $\triangle ABC$ given that $m\angle C = 41^\circ$, $AC = 55$, and $BC = 73$.

$$A = \frac{1}{2}(55)(73) \sin 41; A \approx 1317 \text{ square units}$$

- b. Find the area of the same triangle given that $m\angle A = 90^\circ$, $AC = 55$, and $AB = 48$.

$$A = \frac{1}{2}(48)(55) = 1320 \text{ square units}$$

- c. Discuss with your partner why the formula $A = \frac{1}{2}bh$ could also be used when given $m\angle A = 90^\circ$, $AC = 55$, and $AB = 48$. Summarize your discussion.

The formula $A = \frac{1}{2}bh$ can be used because this is a right triangle. The two measures given were the legs which represent the base and the height.



Consider jotting down what you hear for question 1f when circulating during this activity. Use that as a starting point for a discussion at the conclusion of this activity.



Have students discuss their answers with multiple partners to come to a finalized consensus among different perspectives and approaches.

7.H3.3 Guided Practice: Area of a Triangle

Component Overview: Students will complete two practice questions with finding the area of a triangle using sine.



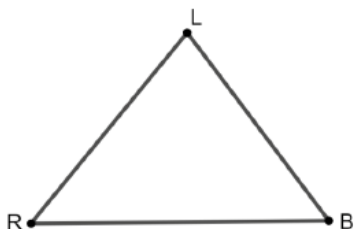
7.H3.3 Guided Practice: Area of a Triangle



The formulas for finding the area of a triangle using sine are:

$$A = \frac{1}{2}bc \sin A \quad A = \frac{1}{2}ac \sin B \quad A = \frac{1}{2}ab \sin C$$

1. $\triangle RLB$ is shown. $RL = 14$, $RB = 16.5$, and $m\angle LRB = 50^\circ$. Find the area of the triangle.



$$A = \frac{1}{2}(16.5)(14) \sin 50; \quad A = 88.478 \text{ square units}$$

2. In $\triangle FPB$, $FP = 9$, $PB = 18$, and $m\angle FPB = 117^\circ$. Find the area of the triangle.

$$A = \frac{1}{2}(9)(18) \sin 117; \quad A = 72.172 \text{ square units}$$



Consider leveraging the “Take Turns” instructional routine during the Guided Practice. This way, students can take turns in the work of the activity, being able to explain the rationale for their choices, and listen to another’s rationale to deepen their understanding.



- How can you determine the best method for finding the area in question 2?
- Why would trigonometry be required to find the area?

7.H3.4 Your Turn!

Component Overview: Students will complete two practice questions with finding the area of a triangle using trigonometry.



7.H3.4 Your Turn!

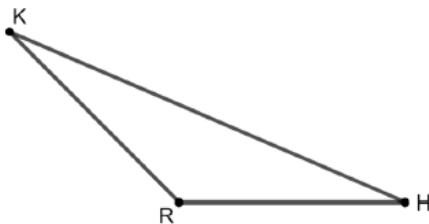
1. In $\triangle NBR$, $NB = 20$, $NR = 18.6$, and $m\angle RNB = 43^\circ$. Find the area of the triangle.

$$A = \frac{1}{2}(20)(18.6) \sin 43; \quad A = 126.852 \text{ square units}$$



Consider using this time to pull together a small group of students who may have been struggling during the Exploration and/or the Guided Practice of this lesson to provide additional support with these questions in a small group setting.

2. $\triangle KRH$ is shown. $KR = 18$, $RH = 16$, and $m\angle KRH = 135^\circ$. Find the area of the triangle.



$$A = \frac{1}{2}(18)(16) \sin 135; \quad A = 101.823 \text{ square units}$$



What is the relationship between finding the area of a triangle with knowing the side values, and finding the area of a triangle using trigonometry?

7.H3.5 Wrap-Up: Cool Down

Component Overview: Students write a short summary of their learning in the lesson. Student responses can be used to identify misconceptions and plan for upcoming instruction.



7.H3.5 Wrap-Up: Cool Down

1. What are the similarities and differences between finding the area of a triangle and finding the area of a triangle using the sine ratio?

Answers will vary.



Consider our online resources for additional enrichment opportunities for students to practice or test their abilities.

ISBN 978-1-952882-74-6



ISBN 978-1-955898-75-1



G111

